CSA4008 - APPLIED MACHINE LEARNING

Presented By

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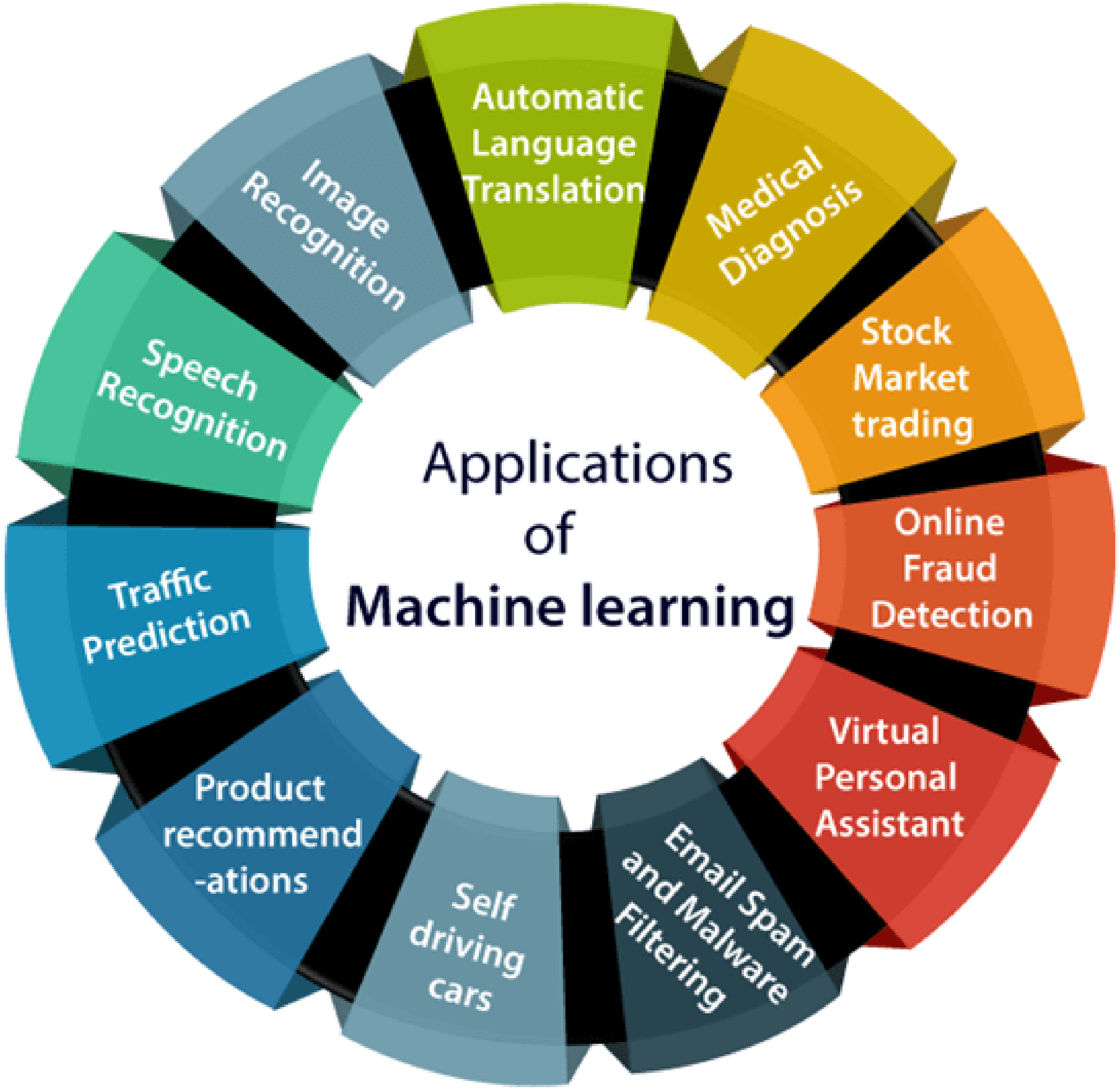
What is Machine Learning?

- Machine Learning is the study of computer algorithms that can improve automatically through experience and by the use of data.
- It is seen as a part of artificial intelligence.
- Machine learning algorithms build a model based on sample data, known as training data, in order to make predictions or decisions without being explicitly programmed.
- Machine Learning is the most popular technique of predicting the future or classifying information to help people in making necessary decisions.
- Machine Learning algorithms are trained over instances or examples through which they learn from past experiences and also analyze the historical data.

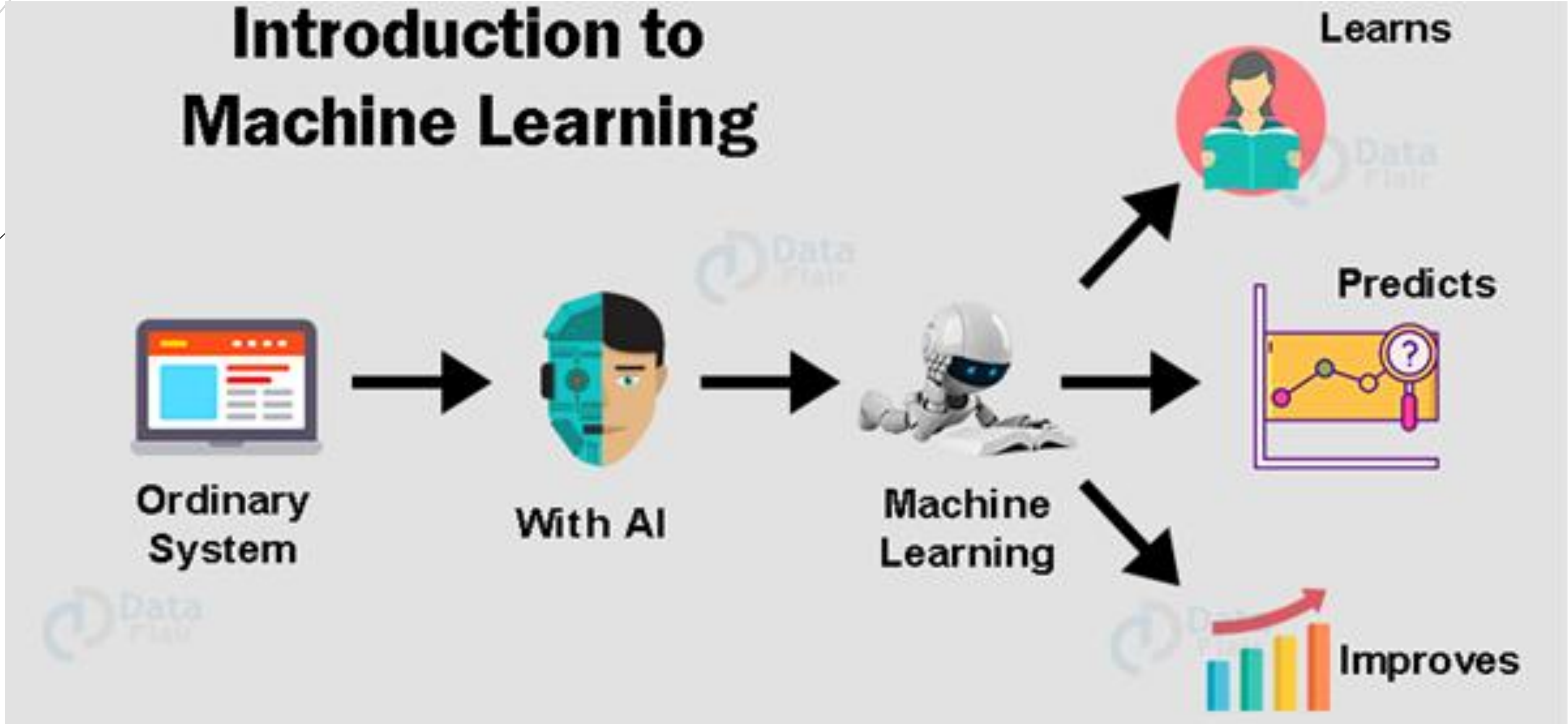
Difference Between AI & ML

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)
Definition	AI is the broader concept of machines being able to carry out tasks in a way that mimics human intelligence.	ML is a subset of AI that enables machines to learn from data and improve over time without being explicitly programmed.
Goal	To create intelligent systems that can simulate human thinking and behavior.	To allow systems to learn from data and make decisions or predictions.
Scope	Broad – includes reasoning, problem-solving, perception, language understanding, and learning.	Narrower – focuses only on learning from data and patterns.
Examples	Chatbots, self-driving cars, facial recognition, game-playing AI (like AlphaGo).	Spam filters, recommendation systems, speech recognition, predictive analytics.
Human Involvement	May require logic and rules set by humans in addition to learning.	Relies heavily on data and algorithms to learn patterns automatically.
Types	Includes ML, Deep Learning, Expert Systems, Robotics, NLP, etc.	Includes Supervised, Unsupervised, Semi-supervised, and Reinforcement Learning.
Learning Capability	Can use learning (ML) or rule-based systems.	Focused entirely on data-driven learning.
Output	Can be decision-making, reasoning, perception, or control.	Predictions, classifications, or pattern discovery.

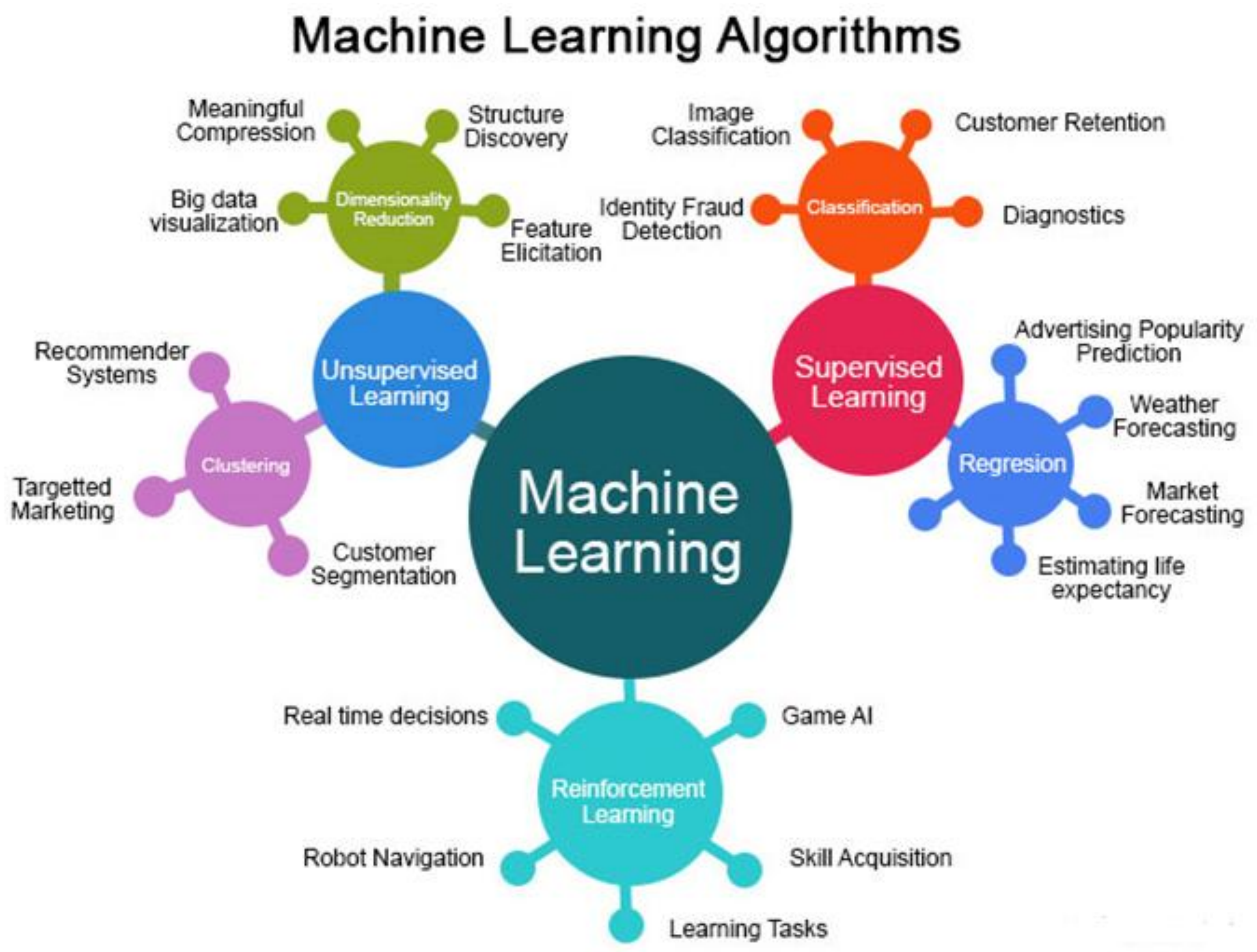
Application of ML

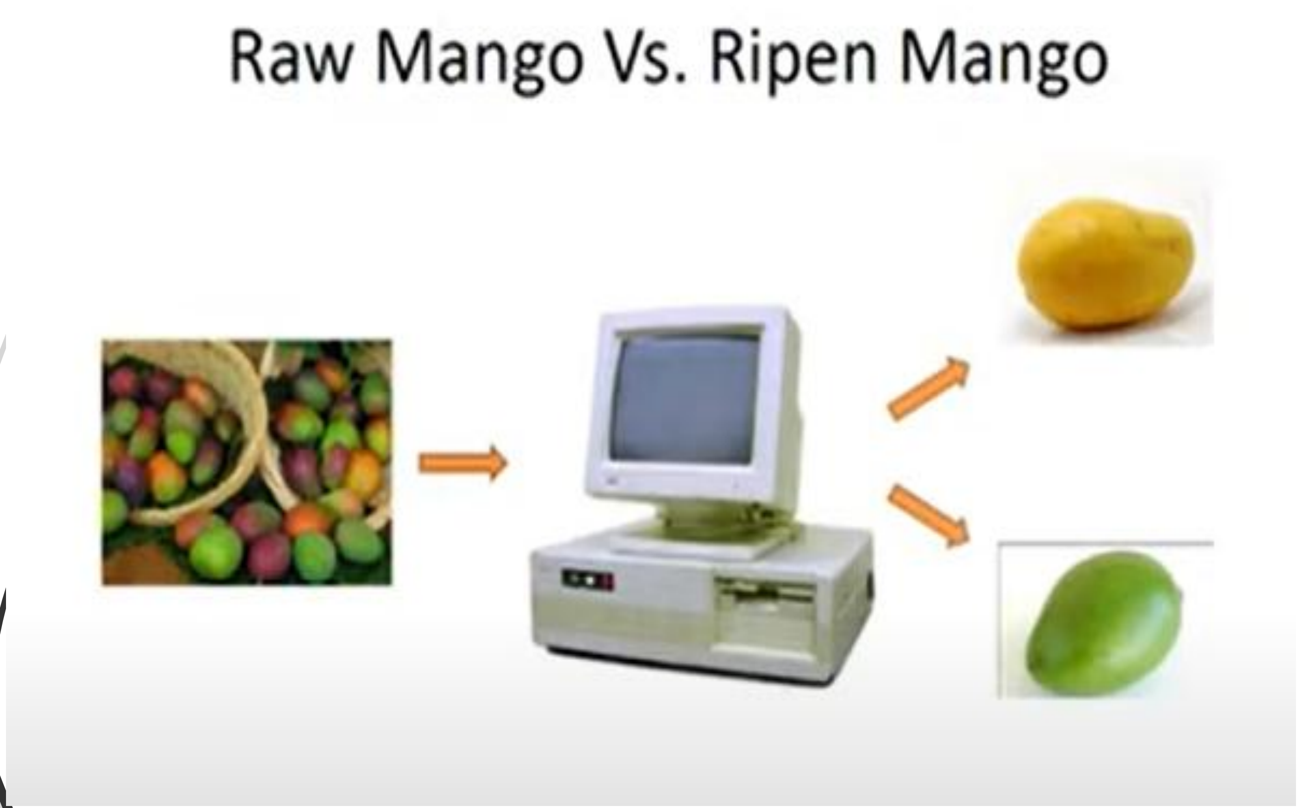


Introduction to Machine Learning



ML Algorithms





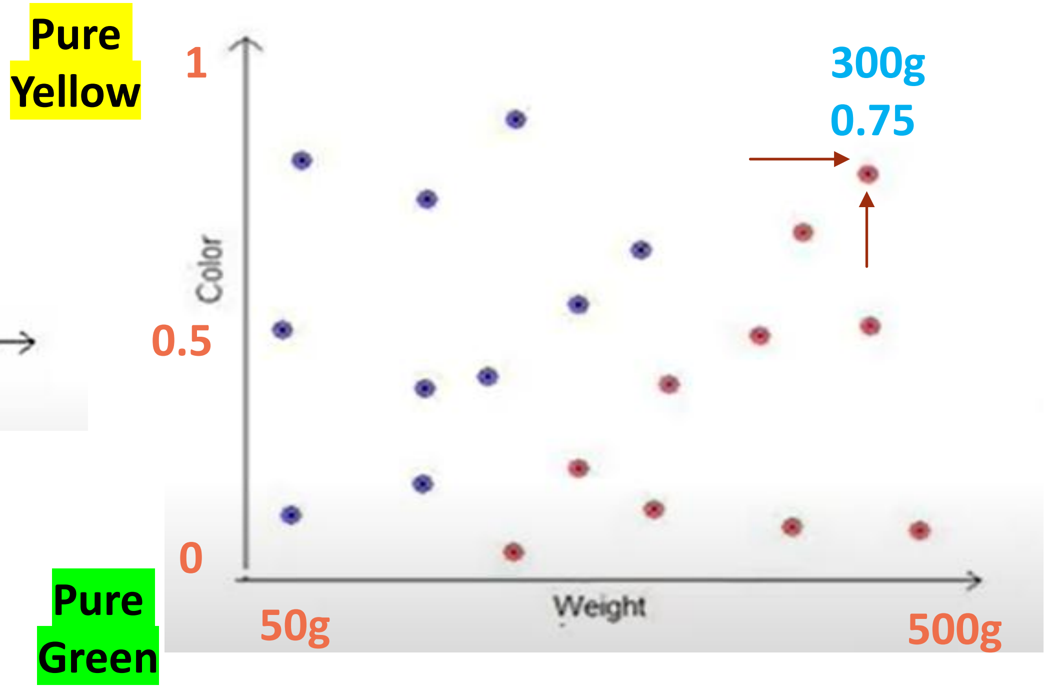
Features:

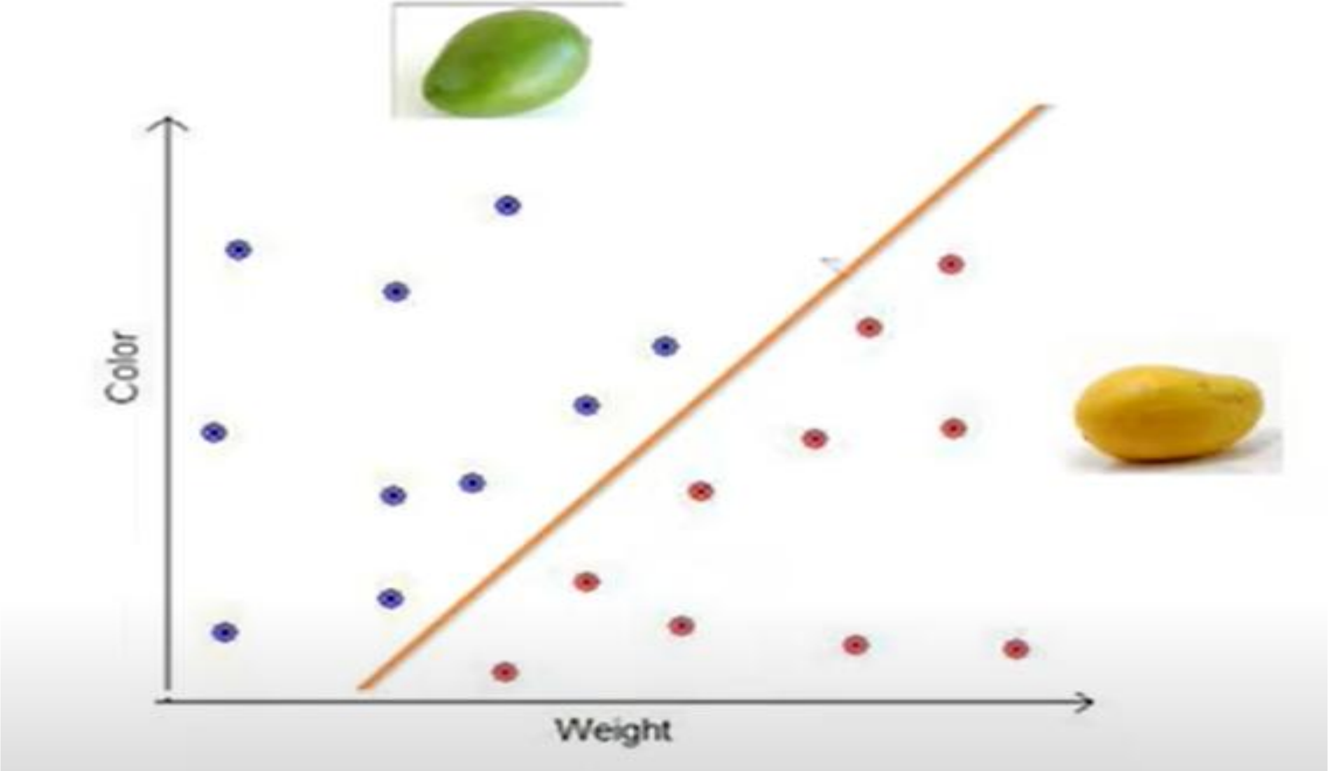
1. **Weight**
2. **Colour**

Red: Ripen Mango

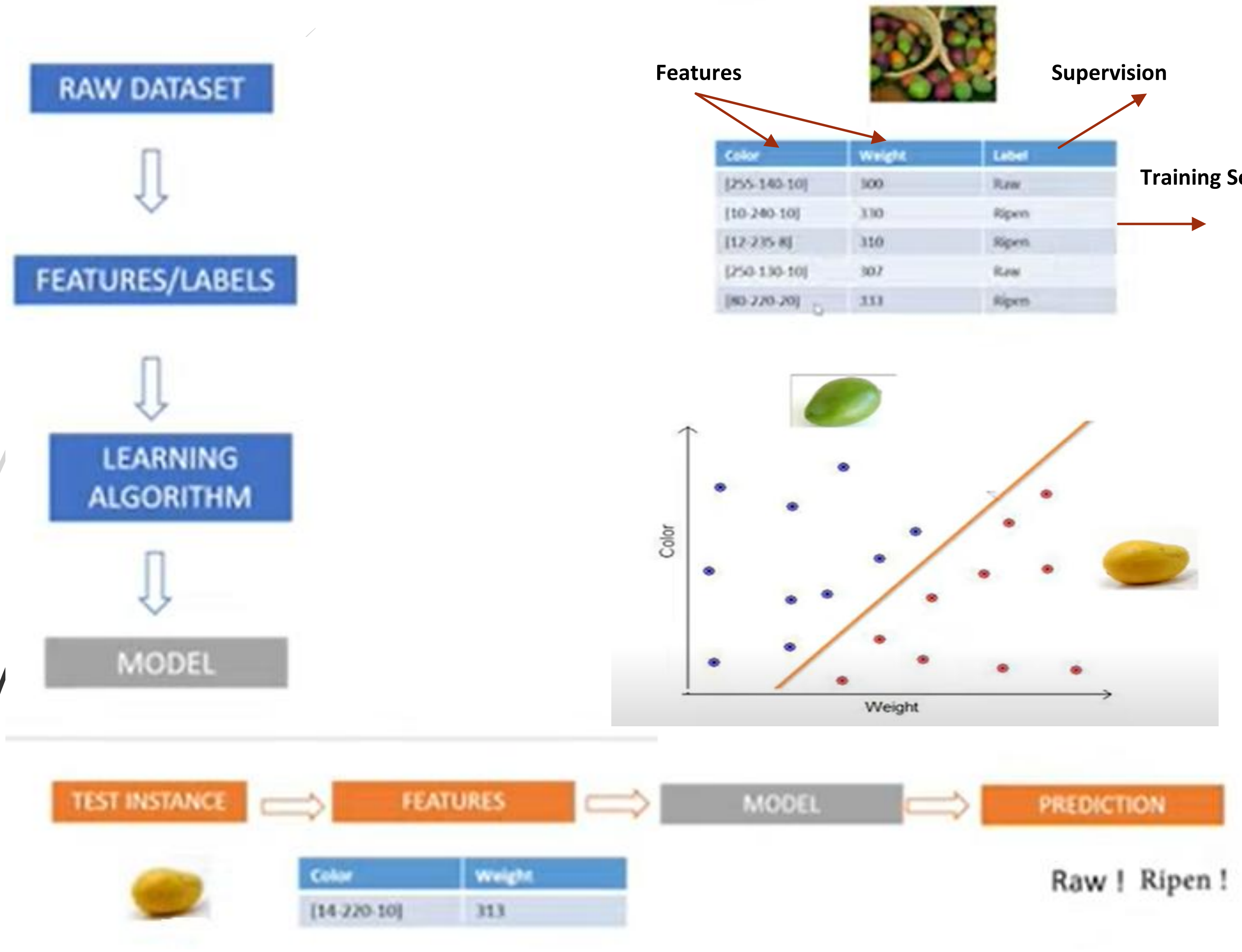
Blue: Raw


Weight
50g
500g


Pure Yellow
1
0.5
0
Pure Green
Color
Weight
50g
500g
300g
0.75


Color
Weight

Supervised Learning

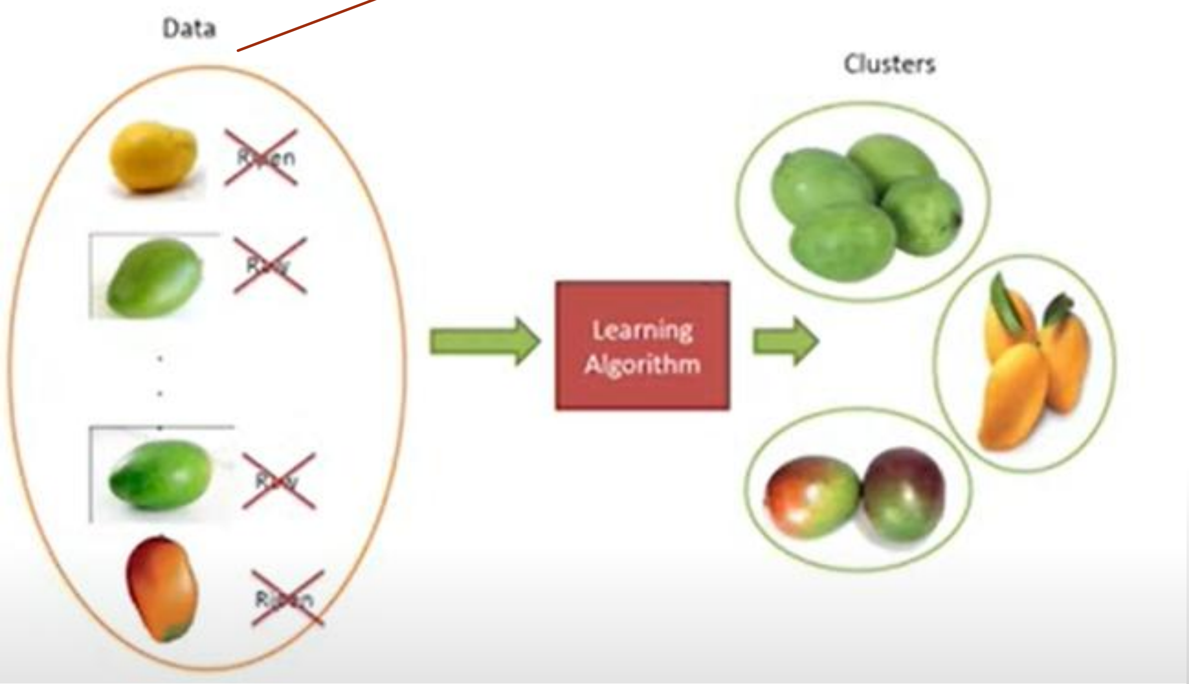


Color	Weight	Label
[255-140-10]	300	Raw
[10-240-10]	330	Ripen
[12-235-8]	310	Ripen
[250-130-10]	307	Raw
[80-220-20]	333	Ripen

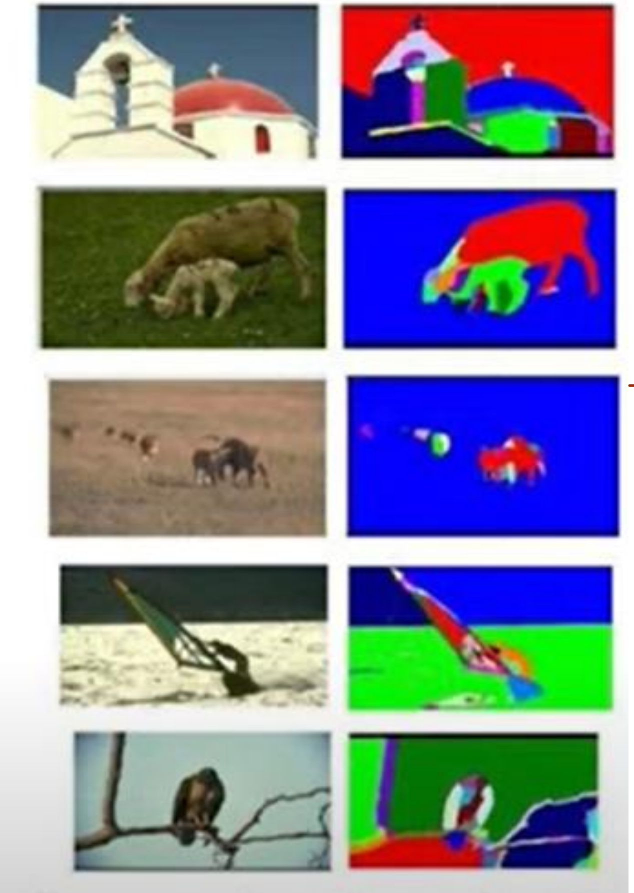
Color	Weight
[14-220-10]	313

Unsupervised Learning

No supervision



Foreground
Background



Segmentation

Find Supervised or Unsupervised

- Data is provided with labels
- Data is grouped into clusters (clustering)
- Input data comes with corresponding output values
- Classifying objects into specific categories (Classification)
- Discovering patterns within the data
- Classifying emails as "spam / not spam"
- Grouping customers into clusters (Customer segmentation)
- Predicting house prices (Prediction)
- Grouping unlabeled data into clusters

- Data is provided with labels - **Supervised**
- Data is grouped into clusters (clustering) - **Unsupervised**
- Input data comes with corresponding output values- **Supervised**
- Classifying objects into specific categories (Classification)- **Supervised**
- Discovering patterns within the data- **Unsupervised**
- Classifying emails as "spam / not spam"- **Supervised**
- Grouping customers into clusters (Customer segmentation)- **Unsupervised**
- Predicting house prices (Prediction)- **Supervised**
- Grouping unlabeled data into clusters- **Unsupervised**

Guess the type of Learning?

- Given a patient's X-ray image, diagnose if he has cancer.

  Ans:

- Automatically group your personal collection of photographs in phone into categories.

  Ans:

- 2. Given a bank customer's profile, should I sanction him a loan?

  Ans:

- 2. Given a audio track, separate the singer's voice from the background music.

  Ans:

Guess the type of Learning?

- Given a patient's X-ray image, diagnose if he has cancer.

  Ans: Supervised Learning

- Automatically group your personal collection of photographs in phone into categories.

  Ans: Unsupervised Learning

- 2. Given a bank customer's profile, should I sanction him a loan?

  Ans: Supervised Learning

- 2. Given a audio track, separate the singer's voice from the background music.

  Ans: Unsupervised Learning

Types of supervised learning

1. Binary Classification

Spam Filter:

1. Spam Mail
2. Not Spam Email

Label

2. Multiclass Classification

Labels = {0,1,2,3,4,5,6,7,8,9}

Cont…

3. Regression

Training Data

Features

Label

	APPLE	ORANGE	GRAPE	AMOUNT
DAY 1	3	2	1	105
DAY 2	1	2	3	112
DAY 3	2	2	1	86
DAY 4	1	1	1	54.50
DAY 5	3	2	1	104
DAY 6	3	3	1	116
DAY 7	3	2	2	???

Real Number= infinite

Test Data

- We are given a table showing quantities of **Apple**, **Orange**, and **Grape** bought on different days and their total **Amount**. We are to determine the amount for **Day 7**, where:
- Apple: 3
- Orange: 2
- Grape: 2
- Amount: ???

- Let's assume the prices are:
- Apple = A
- Orange = O
- Grape = G
- We can set up equations based on the given data:

From the table:

1. `3A + 2O + 1G = 105`
2. `1A + 2O + 3G = 112`
3. `2A + 2O + 1G = 86`
4. `1A + 1O + 1G = 54.5`
5. `3A + 2O + 1G = 104`
6. `3A + 3O + 1G = 116`

- We notice equations 1 and 5 have the same fruit quantities, yet different totals (105 vs 104). So maybe a slight inconsistency exists or prices changed on a day, but let's try with the consistent equations.

Let's take:

- Equation 2: `A + 2O + 3G = 112`
- Equation 3: `2A + 2O + G = 86`
- Equation 4: `A + O + G = 54.5`

Now solve the system.

Step 1: From Eq(4):

`A + O + G = 54.5` → (i)

Step 2: Subtract (i) from Eq(3):

```
2A + 2O + G = 86
- (A + O + G = 54.5)
```

`A + O = 31.5` → (ii)

Step 3: Subtract (i) from Eq(2):

```
A + 2O + 3G = 112
- (A + O + G = 54.5)
```

`O + 2G = 57.5` → (iii)

From (ii): `A = 31.5 - O`

Substitute into (i):

`(31.5 - O) + O + G = 54.5` → `31.5 + G = 54.5` → `G = 23`

Now from (iii):

`O + 2×23 = 57.5` → `O = 11.5`

Now from (ii):

`A + 11.5 = 31.5` → `A = 20`

Final prices:

- Apple (A) = 20
- Orange (O) = 11.5
- Grape (G) = 23

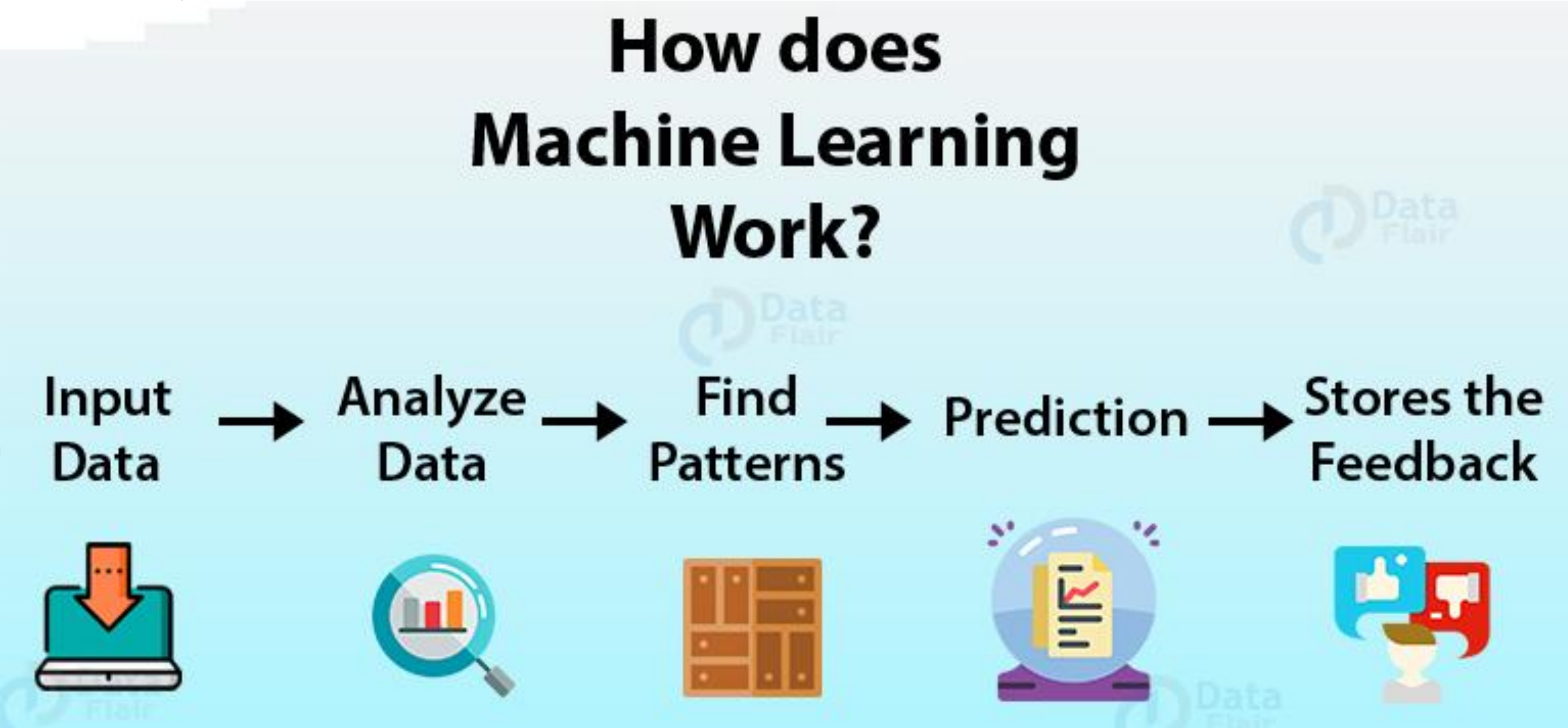
Day 7 values:

- 3 Apples = $3 \times 20 = 60$
- 2 Oranges = $2 \times 11.5 = 23$
- 2 Grapes = $2 \times 23 = 46$

Total = 60 + 23 + 46 = 129

Final Answer: **₹129**

How does Machine Learning Works?

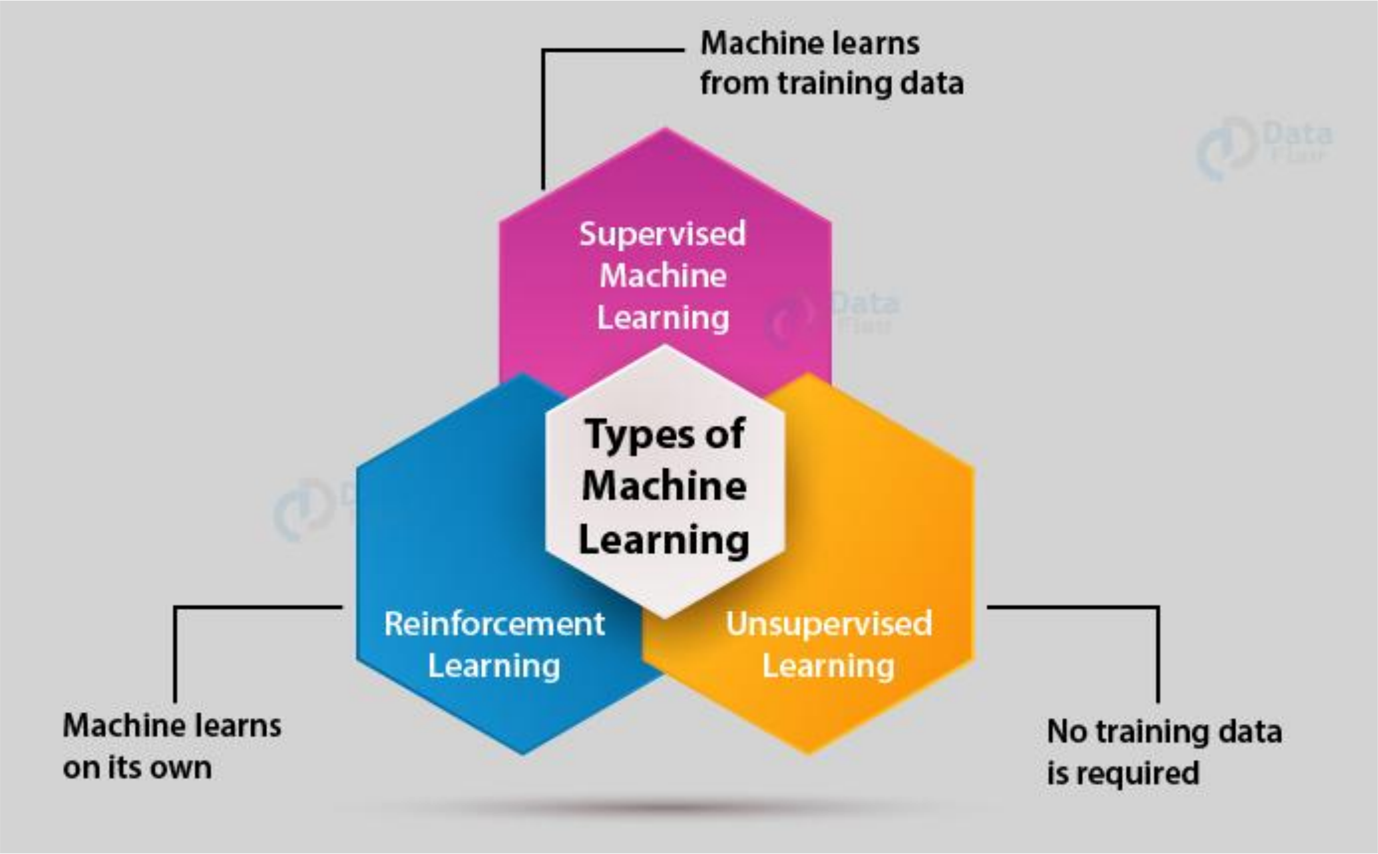


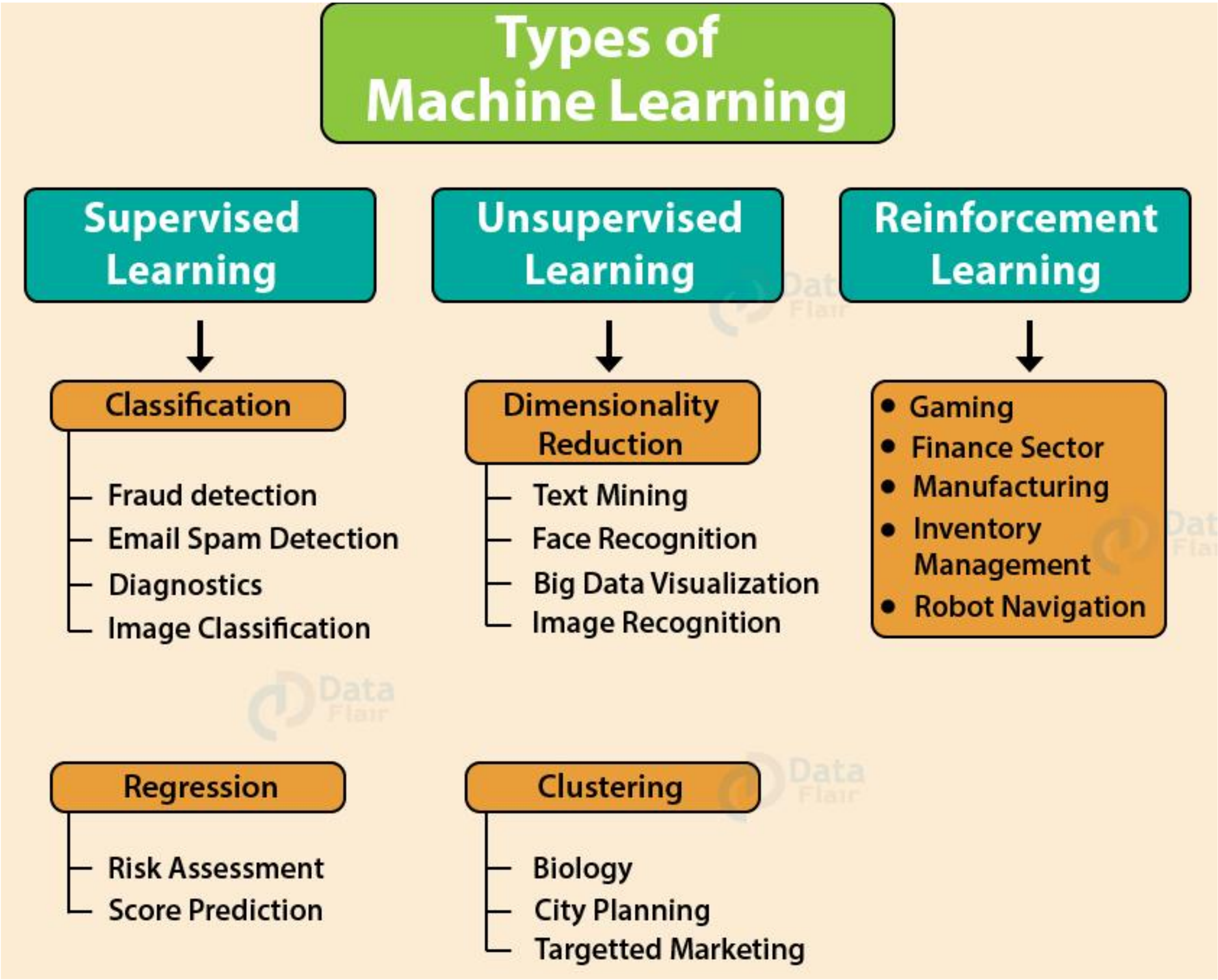
Final Answer: ₹129

Types of Machine Learning

Machine Learning Algorithms can be classified into 3 types:

- **Supervised Learning**
- **Unsupervised Learning**
- **Reinforcement Learning**





Types of Machine Learning
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Classification
Fraud detection
Email Spam Detection
Diagnostics
Image Classification
Dimensionality Reduction
Text Mining
Face Recognition
Big Data Visualization
Image Recognition
• Gaming
• Finance Sector
• Manufacturing
• Inventory Management
• Robot Navigation
Regression
Risk Assessment
Score Prediction
Clustering
Biology
City Planning
Targetted Marketing

Supervised learning

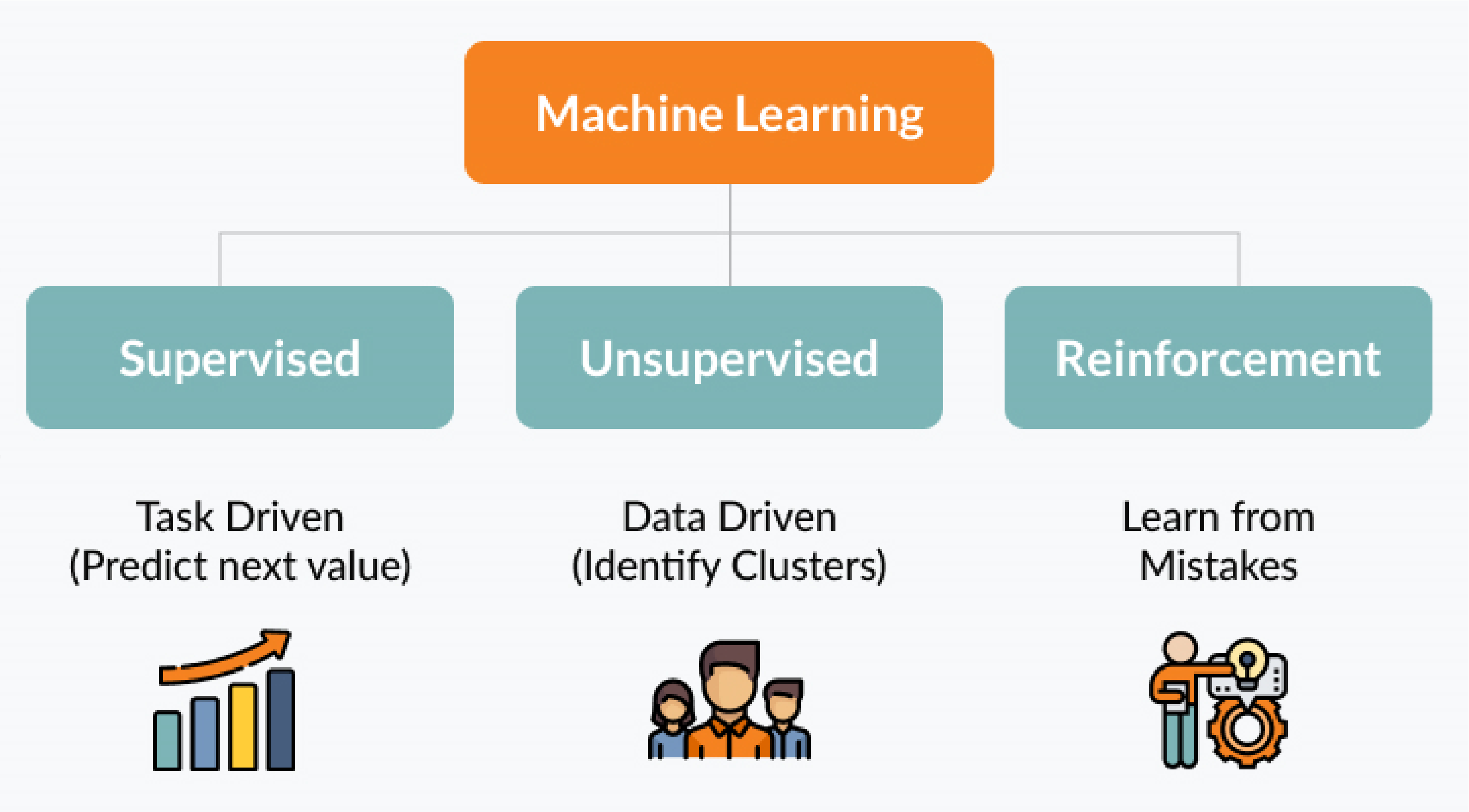
- Supervised learning is that the machine learning task of learning a function that maps an **input** to an **output**.
- The dataset on which we train our model is **labeled**. There is a clear and **distinct mapping** of input and output. Based on the example inputs, the model is able to get **trained** in the **instances**.
- An example of supervised learning is **spam filtering**.
- Based on the **labeled data**, the model is able to determine if the data is **spam** or **not.** This is an easier form of **training**.

Unsupervised Learning

- Unsupervised Learning may be a machine learning technique during which the users don’t got to **supervise the model**. It allows the model to figure on its own to get **patterns** and **knowledge** that was **previously undetected**. It mainly deals with the **unlabeled data**.
- In Unsupervised Learning, there is no labeled data. The algorithm identifies the **patterns** within the **dataset** and **learns** them. The algorithm groups the data into **various clusters** based on their **density.** Using it, one can perform **visualization** on **high dimensional data**.
- One example of this type of Machine learning algorithm is the **Principle Component Analysis**.

Reinforcement Learning

- Reinforcement Learning is an **emerging** and **most popular** type of Machine Learning Algorithm. It is used in various **autonomous systems** like **cars** and **industrial robotics**. The aim of this algorithm is to reach a goal in a **dynamic environment**. It can reach this **goal** based on several rewards that are provided to it by the system.
- It is most heavily used in **programming robots** to perform **autonomous actions**. It is also used in making **intelligent self-driving cars**.
- Let us consider the case of **robotic navigation**.
- The **efficiency** can be improved with further **experimentation** with the agent in its environment. This the main principle behind **reinforcement learning**.


Machine Learning
Supervised
Unsupervised
Reinforcement
Task Driven
(Predict next value)
Data Driven
(Identify Clusters)
Learn from
Mistakes

Machine Learning Algorithms

- **Regression**

Regression models are used **extensively** to **predict values** based on the **variables** that are **dependent** on **several factors**.

- **Decision Tree Learning**

Decision Trees are a **supervised type** of **machine learning algorithms**. These trees are mainly used for **predictive modeling**. We **create** a **decision tree** that is able to **take decisions** based on user input.

Cont...

- **Support Vector Machines**

Support Vector Machines or SVMs are machine learning algorithms that are used to classify data into two categories or classes.

- **Association Rule Learning**

Association Rule Mining is used for finding relationships between several variables that are present in the database. It is a type of data mining technique through which you can discover association between several items.

Cont…

- **Artificial Neural Networks (ANN)**

An Artificial Neural Network is an **advanced form** of **machine learning technique**. These **neural networks** are modeled after the **human nervous system** and are therefore called **neural networks**.

- **Inductive Logic Programming**

In this, logic programming forms the core part to produce a **rule-like learning model**.

Inductive Logic Programming presents the input information, **hypothesis** as well as the **background contextual knowledge** in the form of several rules that have to be followed with **logic**.

Cont…

Reinforcement Learning

The aim of Reinforcement Learning is to **direct** the agent towards **maximizing rewards** and **reach its goal**. This takes place in a **dynamic environment** where the agent has to chart its way to the goal through a **series of trials** and **errors**.

Clustering

In **clustering**, the observations are **divided into groups** or **clusters**. These clusters are formed based on **similar data** and have **similar criteria**. These criteria can be **density** or **similar structure** of the **data**.

There are several clustering techniques that make use of **different criteria** to **cluster the data**.

Cont…

Similarity and Metric Learning

Similarity determination is one of the **key functions** of machine learning. In this form of learning, the **ML model** is provided a **mix of similar** as well as **dissimilar data objects**.

The machine learning model **learns to map** similar objects together and learns a similarity function that allows it to group similar objects together in the future.

Cont…

Bayesian Networks

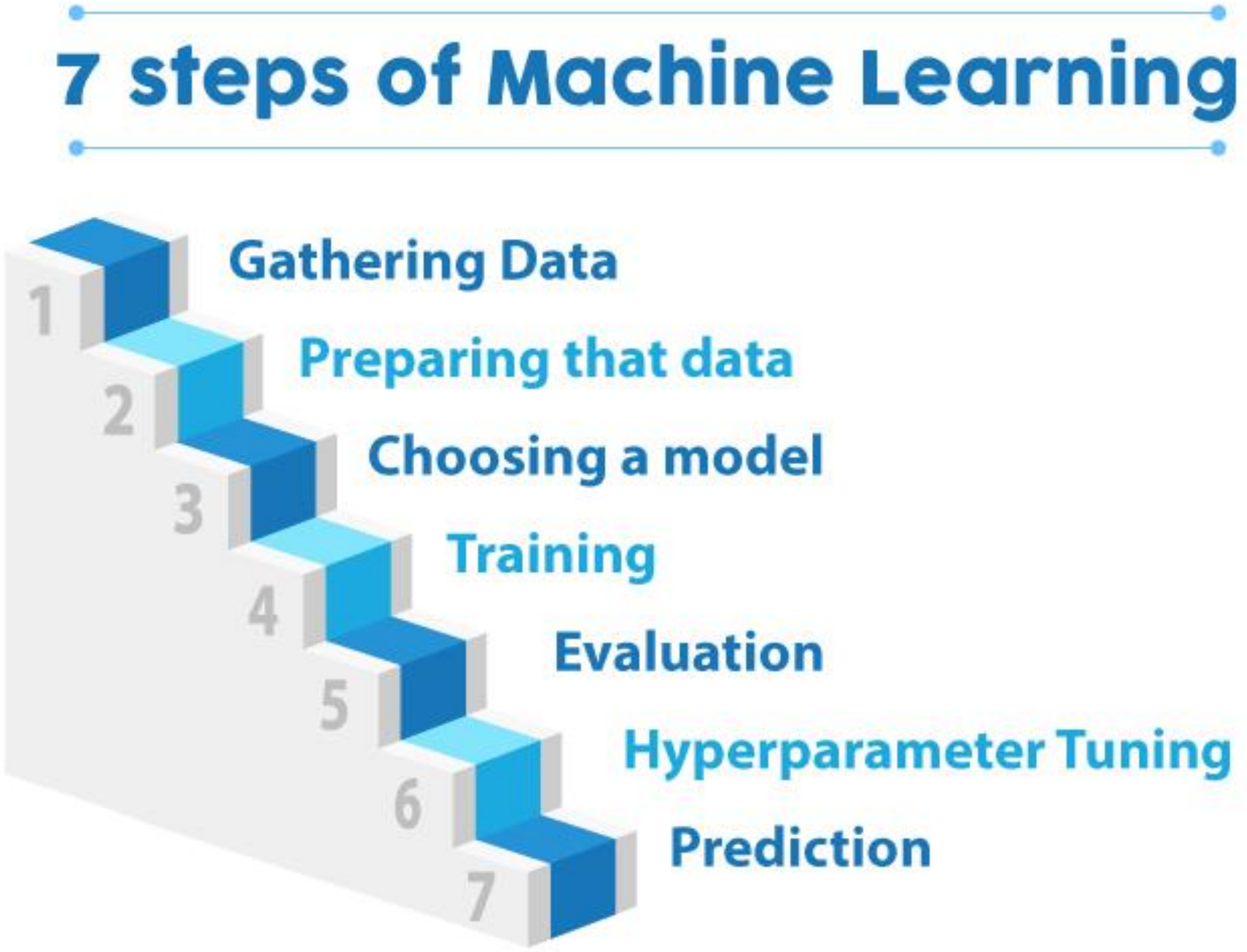
A Bayesian Network is an **acyclic directed graphical model**. This model is also called **DAG** which represents the probability of several **independent conditioned variables**.

More advanced forms of **Bayesian Networks** are **Deep Bayesian Networks**.

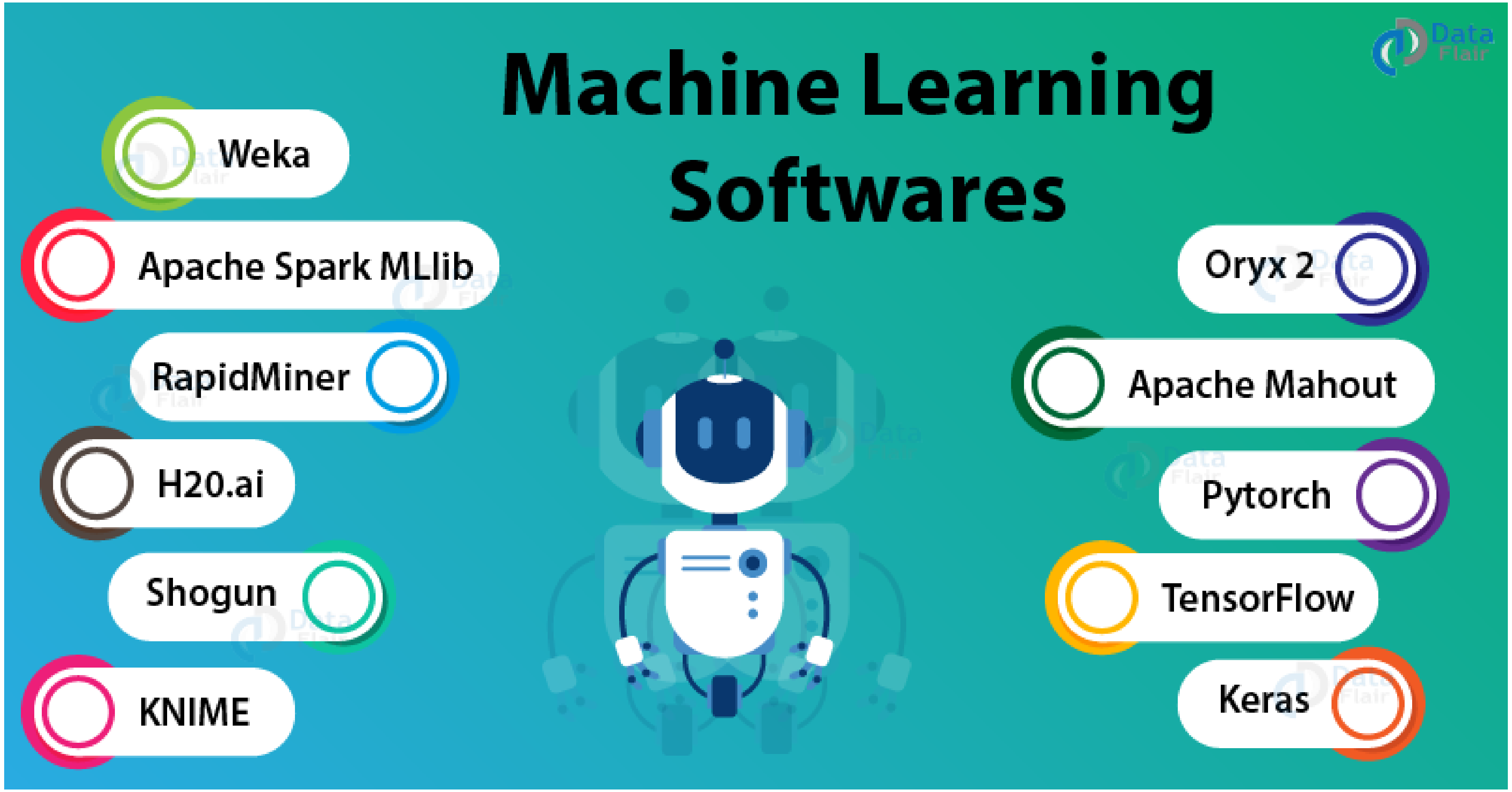
Representation Learning

In order to represent the data in a **more structured format**, we make use of **representation learning**. This formats the **data efficiently** so that the **model** can **train better** to **provide accurate results**.

The representation of data is one of the key factors that can affect the **performance** of the machine learning method. This allows the algorithm to **learn better** from the **data**.

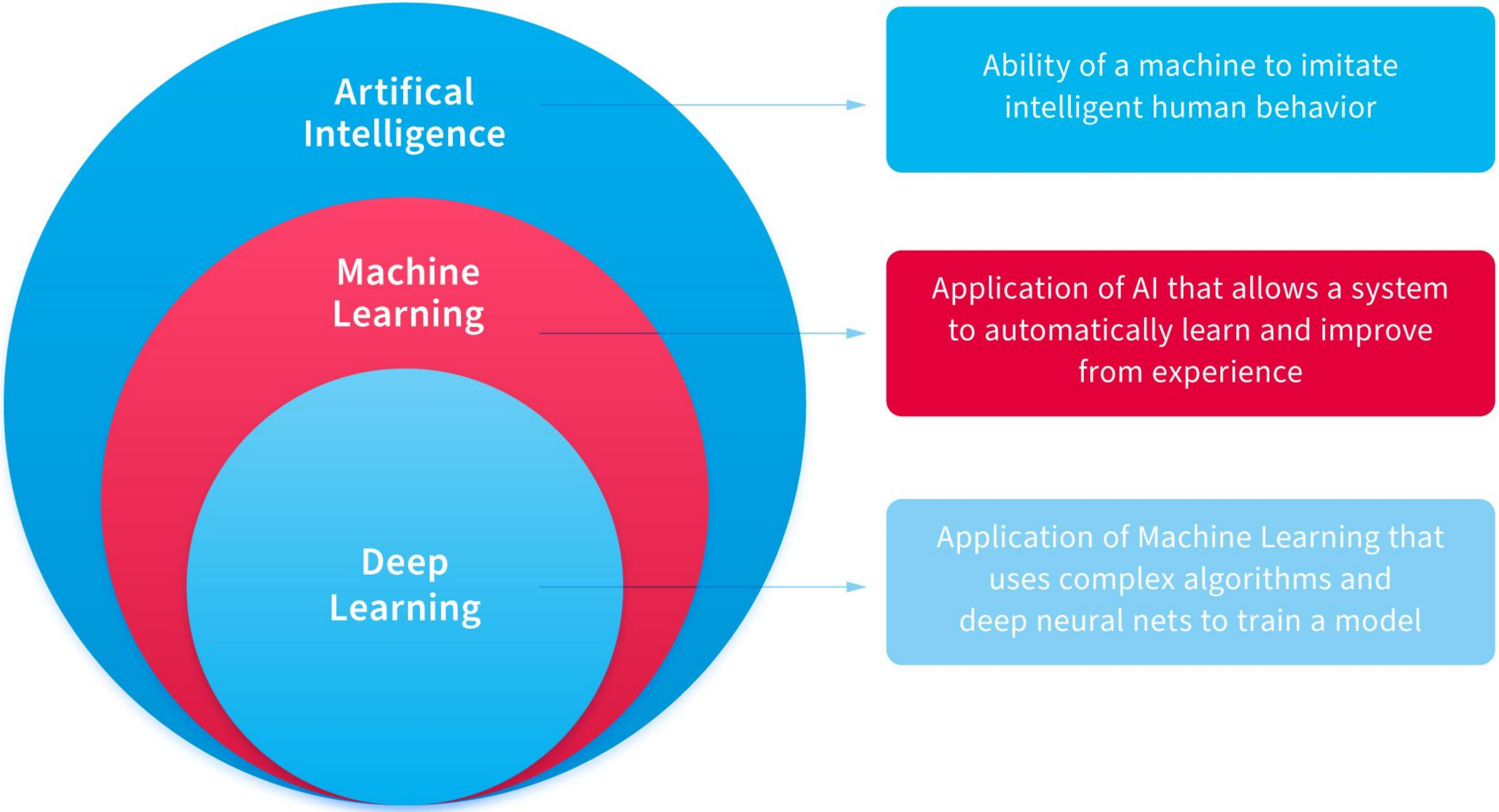

7 steps of Machine Learning
1 Gathering Data
2 Preparing that data
3 Choosing a model
4 Training
5 Evaluation
6 Hyperparameter Tuning
7 Prediction

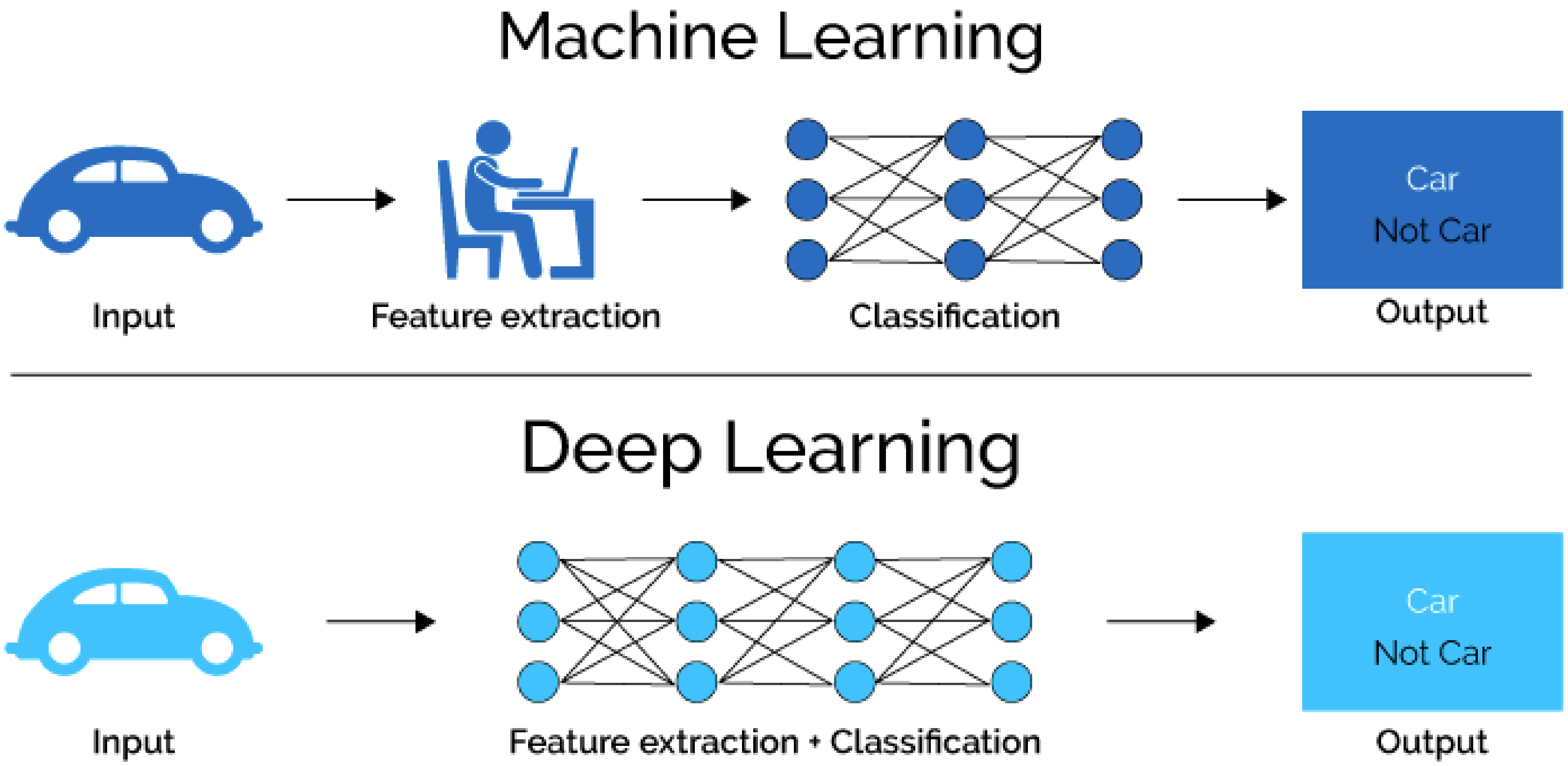
ML Softwares



Traditional Programming

Machine Learning


Artifical Intelligence
Machine Learning
Deep Learning
Ability of a machine to imitate intelligent human behavior
Application of AI that allows a system to automatically learn and improve from experience
Application of Machine Learning that uses complex algorithms and deep neural nets to train a model


Machine Learning
Car
Not Car
Input
Feature extraction
Classification
Output
Deep Learning
Car
Not Car
Input
Feature extraction + Classification
Output

Theory of learning

- Machine Learning Theory, also known as **Computational Learning Theory**, aims to understand the fundamental principles of learning as a computational process and combines tools from Computer Science and Statistics.

Feasibility of Learning

- Feasibility studies are common across **industries and disciplines**.
- They are an important project planning tool that can help you identify points of failure in a project before any money or time gets invested.

Example #1 – Expansion of Hospital

- If the hospital wants to expand its area of a building, then it may conduct the feasibility study, which will help it in determining whether a hospital should go ahead with the project of expansion or not.


Feasibility Study Examples
Feasibility Study on the Expansion of Hospital
Example 1
Feasibility Study on Starting a New Family Restaurant
Example 2
Feasibility Study on the Expansion of School
Example 3

Feasibility of Learning

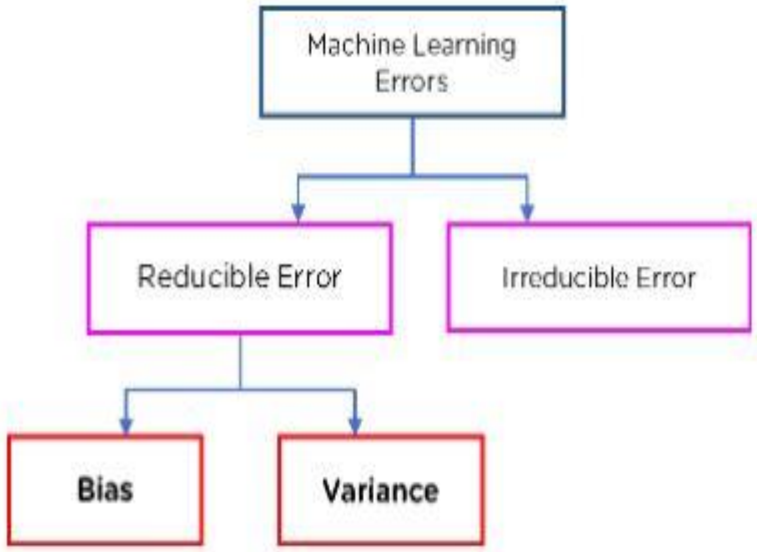
- *There are four main elements that go into a feasibility study:*
- Technical feasibility,
- Financial feasibility,
- Market feasibility (or market fit) and
- Operational feasibility.

Error and noise

- The errors are referred **to as noise**.
- Data noise in machine learning can cause problems since the algorithm interprets the noise as a pattern and can start generalizing from it.
- As a result, by using an algorithm, any data scientist must deal with the noise in data science.

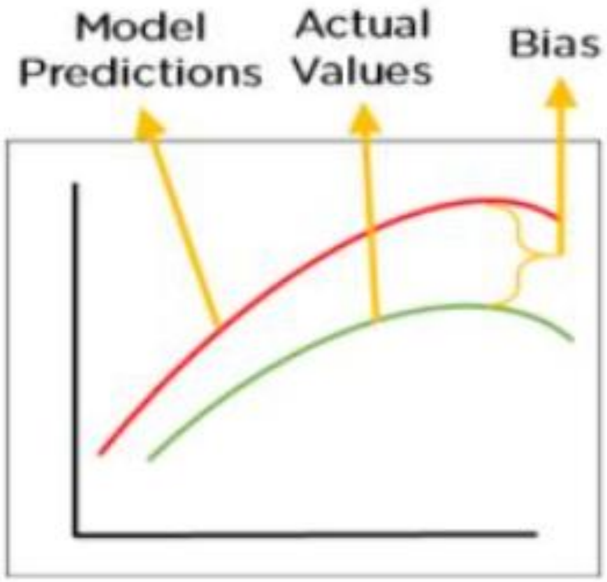
What are the errors in ML?

- There are two main types of errors present in any machine learning model.
- They are **Reducible Errors and Irreducible Errors.**
- Irreducible errors are errors which will always be present in a machine learning model, because of unknown variables, and whose values cannot be reduced.



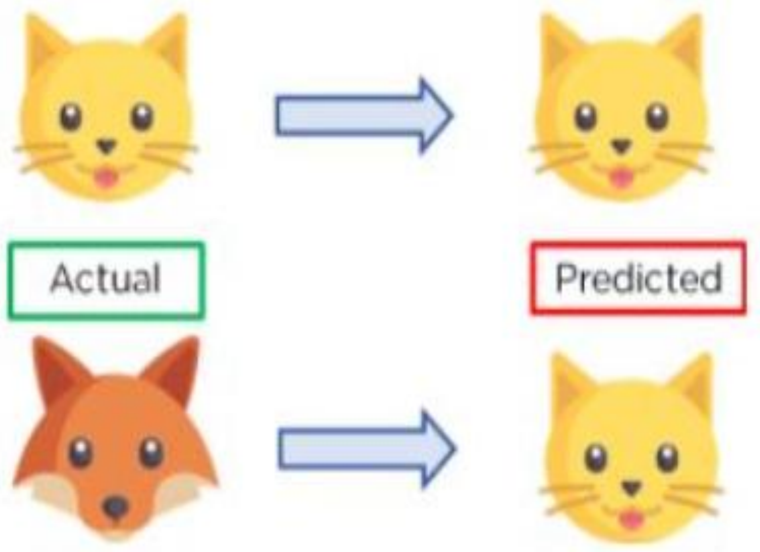
What is Bias?

- Bias is the difference between our actual and predicted values.
- Bias is the simple assumptions that our model makes about our data to be able to predict new data.



What is Variance?

- Variance is the very opposite of Bias. During training, it allows our model to ‘see’ the data a certain number of times to find patterns in it.
- If it does not work on the data for long enough, it will not find patterns and bias occurs.
- We can define variance as the model’s sensitivity to fluctuations in the data. Our model may learn from noise.



Difference Between Bias and Variance

Aspect	Bias	Variance
What is it?	Error caused when the model is **too simple** and makes overly strong assumptions.	Error caused when the model is **too sensitive** to small changes in training data.
Cause	Model ignores important patterns and learns only simple rules (**Underfitting**).	Model learns even noise and unnecessary patterns (**Overfitting**).
Effect on Model	Fails to capture the true relationship.	Captures noise as if it were signal; becomes inconsistent.
Training Error	**High**	**Low**
Testing Error	**High**	**High** (because of inconsistency)
Example Behavior	Always predicts around the average value, regardless of input.	Predictions change a lot even for small data changes.
Ideal Case	Low Bias + Low Variance = Good Generalization.	Low Bias + Low Variance = Good Generalization.

Training versus Testing

- Training data trains the model while testing checks (tests) whether this **built model works correctly or not**. However, some users still can use their training data to make predictions.

Train Data

- Train data refers to the dataset used to build a ***machine learning model***. The model learns from the examples in this dataset - that is, it identifies the relationship **between input and output.**
- **Example:**
  Suppose we are building a machine learning model using data from 1000 students.
- **Input features: Daily study hours, class attendance, internal exam marks**
- **Output label: Final exam marks**
- Out of these, data from **800 students** is used as the **training data**. The model learns from these **800 examples to identify patterns and make predictions.**

Test Data

- Test data is used to **evaluate the performance** of the trained model. It contains **new data** the model has not seen during training.
- **Example:**
  The remaining **200 students' data** is used as **test data**.
  Based on their input values (study hours, attendance, etc.), the model tries to **predict** their final exam marks.
  The predicted marks are compared with the actual marks to check how accurately the model performs.

Type	Purpose	Sample Size (Example)
Train Data	To train the model	800 students
Test Data	To test the model's performance	200 students

This kind of data split helps in checking:
•Whether the model **generalizes well** to unseen data
•Whether the model is **overfitting** the training data

Example: Train vs Test Data

Train Data (Model learns pattern):

Study Time (Hours)	Exam Marks
2 Hours	60 Marks
3 Hours	70 Marks
4 Hours	80 Marks

- ***From this, the model learns:***
  - "As study time increases, exam marks increase."
  - Pattern: Every extra hour = +10 marks

Test Data (Model makes prediction):

Study Time (Hours)	Predicted Marks
5 Hours	???

Question:

1. What type of data is the **new student's input** (Train or Test data)?

2. Based on the pattern in the training data, what would be the **predicted marks** for 5 hours of study?

3. Why is it important to test the model on new data?

Answer:

- The new student's data is **Test Data**, because the model has never seen it before and it's being used to check model performance.
- The model observes that for every 1-hour increase in study time, marks increase by 10:
  - 2 hrs → 60, 3 hrs → 70, 4 hrs → 80
  - So, for **5 hours**, the predicted marks = **90**.
- Testing on new data is important to evaluate if the model has **generalized well** — i.e., whether it works not just on training examples but also on real-world or unseen cases. This helps identify **overfitting** or **underfitting**.

Problems

- A machine learning model is being developed to **predict the price of a bicycle** based on its **weight**.
- You are given the following data:

Bicycle Weight (kg)	Price (₹)
10	5000
12	6000
14	7000
16	8000

Questions:

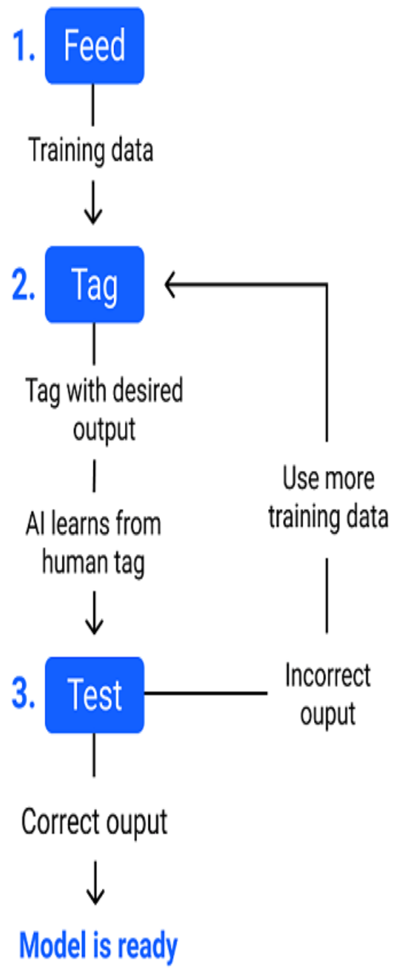
1. Which data from the above table can be used as **Training Data**?
2. If a bicycle weighs **18 kg**, what price will the model likely predict, based on the pattern?
3. What is the **type of data** for the 18 kg bicycle input — Train Data or Test Data?

Answer:

1. The first 4 rows (10, 12, 14, and 16 kg) can be used as **Training Data** — they have known inputs and outputs that help the model learn the relationship between weight and price.
2. The price increases by ₹1000 for every 2 kg increase in weight.
   So, for 18 kg:
   Predicted Price = ₹9000

3. The 18 kg bicycle is **Test Data**, because it is a new, **unseen input** used to test the model's prediction accuracy.

What is the purpose of testing in ML?

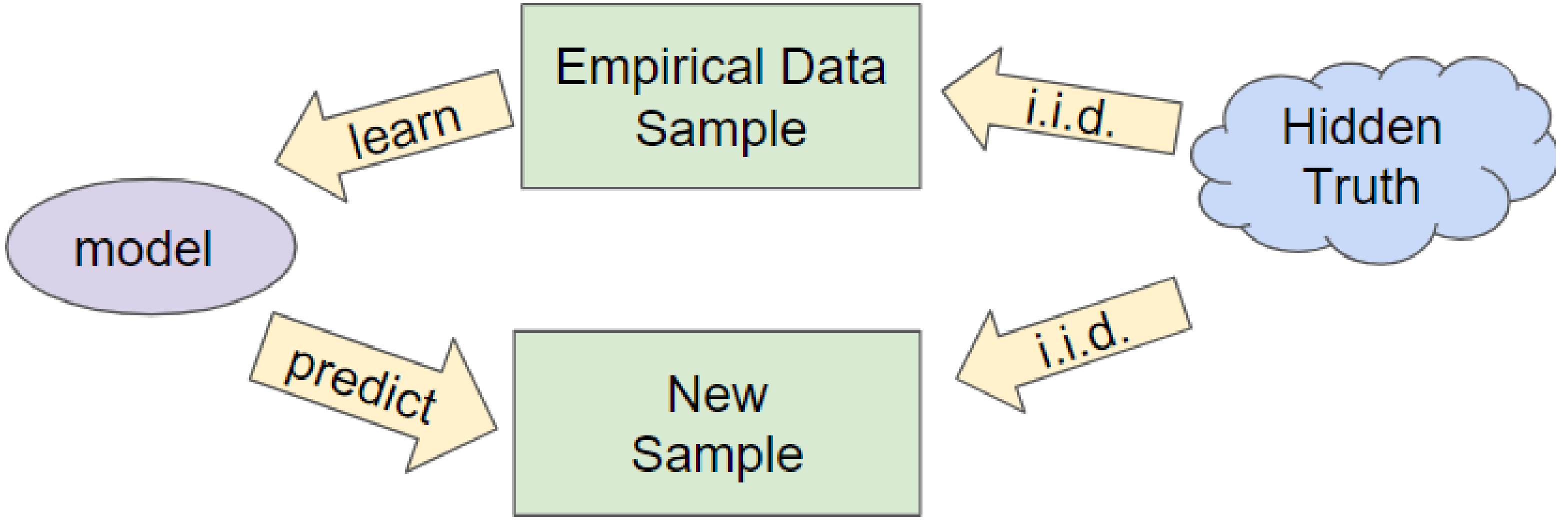


- ML model testing enables data scientists
- to conduct quality assurance of data, features, algorithms, or model parameters to: Eliminate malfunctions and increase robustness: Conducting different tests to assess different aspects of the model enables the root cause of potential problems.

Theory of Generalization

- The generalization of machine learning models is the ability of a model to classify or forecast new data. When we train a model on a dataset, and the model is provided **with new data absent from the trained set**, it may perform well. Such a model is generalizable.

- Generalization refers to your model's ability to adapt **properly to new, previously unseen data, drawn from** the same distribution as the one used to create the model.



- Goal: predict well on new data drawn from (hidden) true distribution.
- Problem: we don't see the truth.
    - We only get to sample from it.

Generalization Bound

- A machine learning model should not only perform well on the **training data** but also on **new, unseen data**. This is called **Generalization**.

- **Generalization Bound** tells us the **mathematical limit on the possible errors** the model might make on new data.

Example

- Imagine you teach a student **10 math problems**. The student solves all **10/10 correctly**.
- Now, if you give the student **10 new problems**, will they also solve them correctly?

This is a **Generalization Question**.

- If the new problems are similar, the student might do well.
- Otherwise, mistakes are possible.
- **Generalization Bound says:**
- "This student might make at most **2 mistakes** on new problems." It gives a **mathematical limit on possible future mistakes.**

Training Error:

- Very low on the training data.
- Sometimes even 0%.

Test Error (Generalization Error):

- Errors happen on new data.
- **Generalization Bound** sets a limit on this error.

- **True Error ≤ Empirical Error + Complexity Term + Confidence Term**
- **Where:**
- **Empirical Error** = Training Error
- **Complexity Term** = How complex the model is
- **Confidence Term** = Our confidence level in this bound holding true.

Type	Training Performance	Testing Performance
Overfitting	Very Good	Poor
Underfitting	Poor	Poor
Good Model	Good	Good

What is the difference between generalization and regularization in ML?

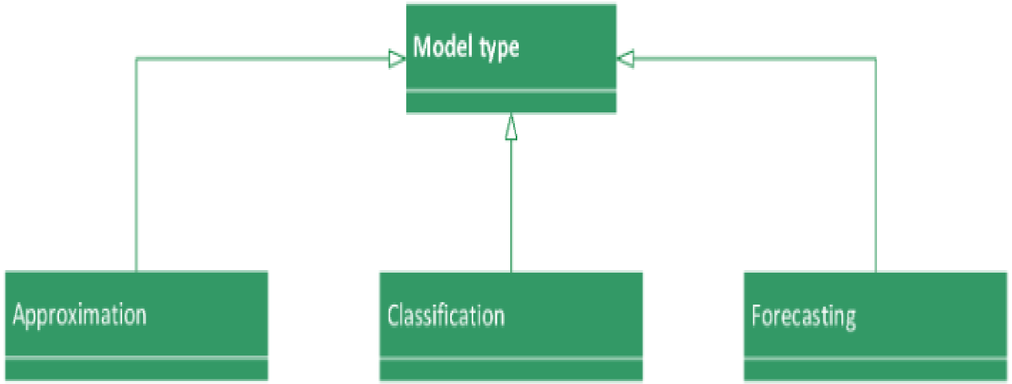
Aspect	Generalization	Regularization
Definition	The ability of a machine learning model to perform well on **unseen data** (data it was not trained on).	A set of **techniques** used during training to improve generalization by reducing overfitting.
Purpose	To ensure the model's performance is consistent on both training and unseen data.	To **penalize complex models** and avoid overfitting to the training data.
Focus	Outcome on new data.	Process during training.
How it's Achieved	Through proper training, sufficient data, and appropriate model complexity.	By adding **penalty terms** like L1, L2, or dropout to the loss function.
Example	A model correctly predicts on a new test dataset.	L1 (Lasso), L2 (Ridge), Dropout, Early Stopping.

Common Techniques to Reduce Overfitting

Techniques	Description
Regularization (L1/L2)	Adds penalty for complex models.
Dropout	Randomly drops neurons during training.
Early Stopping	Stops training when validation error increases.
Cross-Validation	Evaluates model stability on multiple subsets.
Simpler Models	Reduces model complexity (fewer layers, parameters).
More Data	Provides broader examples for learning patterns.
Data Augmentation	Creates variations of training data to improve robustness.

Approximation in ML

- An approximation model assigns a set of input variables to one or several output variables.
- Here the model learns from knowledge represented by a data set consisting of instances with input and target variables.
- The targets are a specification of the response to the inputs.



- In **Machine Learning**, **Approximation** refers to how well a model can **represent the true relationship** between the **input data and the output** labels. Since real-world data is often noisy and complex, ML models aim to **approximate** the actual underlying function rather than model it exactly.

Types of Approximation in ML

Type	Description
Function Approximation	Approximating a mathematical function mapping inputs to outputs (e.g., regression models, neural networks).
Hypothesis Approximation	Selecting the best hypothesis from a set of possible models to approximate the target function.
Model Approximation	Simplifying complex models to reduce computational costs, even if it slightly sacrifices accuracy.
Probabilistic Approximation	Using probabilities to approximate uncertain outputs (e.g., Bayesian methods).

Difference Between Overfitting and Underfitting

S.No	Overfitting	Underfitting
1	The model learns the training data **too well**, including noise and unnecessary patterns.	The model **fails to learn** the underlying pattern in the training data.
2	Very high accuracy on training data, poor performance on new/unseen data.	Poor performance on both training and unseen data.
3	The model is **too complex** (too many features, parameters).	The model is **too simple** (insufficient features, too few parameters).
4	**High training accuracy**, **low testing accuracy**.	Both **training and testing accuracy are low**.
5	**100%** Train Accuracy, **60%** Test Accuracy.	**60%** Train Accuracy, **55%** Test Accuracy.
6	Poor **generalization** to new data.	Model **fails to capture important patterns**.
7	A student memorizes questions and answers but can't answer new questions.	A student doesn't even understand the basics.

Why is Approximation Important in ML?

- Real-world functions are often **unknown or too complex.**
- Exact solutions are often impossible; approximations make the models **practical and generalizable.**
- Helps achieve a **balance between bias and variance.**

Approximation - Generalization Tradeoff in Machine Learning

- In Machine Learning, the **Approximation - Generalization Tradeoff** refers to the balance between:
- How well a model can **fit the training data** (Approximation).
- How well it can **perform on unseen data** (Generalization).
- A **good ML model** must manage both:
- Approximate the underlying function correctly.
- Avoid overfitting the training data to ensure good generalization.

Generalization tradeoff in ML

Aspect	Approximation	Generalization
Focus	Fitting the model to training data.	Performing well on unseen (test) data.
Too much	May lead to overfitting (memorizing training data).	May lead to underfitting (model too simple).
Goal	Reduce training error.	Reduce test error.
Conflict	Complex models approximate better but generalize poorly.	Simple models generalize better but may underfit.

How to Manage the Tradeoff:

Strategy	Impact
Regularization	Controls complexity to improve generalization.
Cross-validation	Helps monitor how well the model generalizes.
Early stopping	Prevents overfitting during training.
Model selection	Choose models appropriate to data complexity.

Problem1: A machine learning model is trained to predict **house prices (₹)** based on the **area in square feet**. Also find the MSE.

Training Data:

Area (x)	Price (y)
1000	200000
1500	250000
2000	300000

Test Data:

Area (x)	Actual Price
1200	220000
1800	280000
2800	380000

Step 1: Compute Intermediate Values

x	y	$x \times y$	x^2
1000	200000	200,000,000	1,000,000
1500	250000	375,000,000	2,250,000
2000	300000	600,000,000	4,000,000

$$\sum x = 4500, \quad \sum y = 750000, \quad \sum xy = 1,175,000,000, \quad \sum x^2 = 7,250,000, \quad n = 3$$

Step 2: Find the Slope (m)

$$m = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

Substitute:

$$m = \frac{3(1,175,000,000) - (4500)(750000)}{3(7,250,000) - (4500)^2}$$

$$m = \frac{3,525,000,000 - 3,375,000,000}{21,750,000 - 20,250,000} = \frac{150,000,000}{1,500,000} = \boxed{100}$$

Step 3: Find the Intercept (b)

$$b = \bar{y} - m \cdot \bar{x}$$

Mean values:

$$\bar{x} = \frac{4500}{3} = 1500, \quad \bar{y} = \frac{750000}{3} = 250000$$

Now:

$$b = 250000 - 100 \cdot 1500 = 250000 - 150000 = \boxed{100000}$$

Step 4: Final Prediction Model

- ***This is the regression equation used to predict house prices based on area.***

$$\boxed{\text{Price} = 100 \cdot \text{Area} + 100000}$$

Step 5: Make Predictions on Test Data

- **Prediction for 1200 sqft:**

$$\text{Price} = 100 \cdot 1200 + 100000 = \boxed{220000}$$

- **Prediction for 1800 sqft:**

$$\text{Price} = 100 \cdot 1800 + 100000 = \boxed{280000}$$

- **Prediction for 2800 sqft:**

$$\text{Price} = 100 \cdot 2800 + 100000 = \boxed{380000}$$

Answer:

Area (sqft)	Actual Price	Predicted Price	Match?
1200	220000	220000	✓
1800	280000	280000	✓
2800	380000	380000	✓

The model generalizes perfectly on the test data - both slope and intercept were correctly derived, and predictions match actual prices.

Step 6: Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = number of samples

A. Training Data Predictions

Area (x)	Actual (y)	Predicted ($\hat{y}$)	Error $y - \hat{y}$	Squared Error
1000	200000	100×1000 + 100000 = 200000	0	0
1500	250000	100×1500 + 100000 = 250000	0	0
2000	300000	100×2000 + 100000 = 300000	0	0

$$\text{Training MSE} = \frac{0+0+0}{3} = \boxed{0}$$

B. Testing Data Predictions

Area (x)	Actual (y)	Predicted ($\hat{y}$)	Error $y - \hat{y}$	Squared Error
1200	220000	100×1200 + 100000 = 220000	0	0
1800	280000	100×1800 + 100000 = 280000	0	0
2800	380000	100×2800 + 100000 = 380000	0	0

$$\text{Testing MSE} = \frac{0+0+0}{3} = \boxed{0}$$

Dataset	MSE
Training Data	0
Testing Data	0

The model perfectly fits both training and testing data — no error at all, so the Mean Squared Error is zero.

Problem2: A machine learning model is trained to predict **house prices (₹)** based on the **area in square feet**. Also find the MSE.

Training Data:

Area (x)	Price (y)
1000	205000
1500	252000
2000	298000

Test Data

Area (x)	Actual Price (y)
1200	220000
1800	275000
2500	345000

Step 1: Compute Intermediate Values

x	y	x × y	x^2
1000	205000	205,000,000	1,000,000
1500	252000	378,000,000	2,250,000
2000	298000	596,000,000	4,000,000

$$\sum x = 4500, \quad \sum y = 755000, \quad \sum xy = 1,179,000,000, \quad \sum x^2 = 7,250,000, \quad n = 3$$

Step 2: Calculate Slope (m)

$$m = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

$$m = \frac{3 \cdot 1{,}179{,}000{,}000 - 4500 \cdot 755000}{3 \cdot 7{,}250{,}000 - 4500^2}$$

$$m = \frac{3,537,000,000 - 3,397,500,000}{21,750,000 - 20,250,000} = \frac{139,500,000}{1,500,000} = \boxed{93}$$

Step 3: Calculate Intercept (b)

$$\bar{x} = \frac{4500}{3} = 1500, \quad \bar{y} = \frac{755000}{3} = 251666.67$$

$$b = \bar{y} - m \cdot \bar{x} = 251666.67 - 93 \cdot 1500 = 251666.67 - 139500 = \boxed{112166.67}$$

Step 4: Final Prediction Model

$$\boxed{\text{Price} = 93 \cdot \text{Area} + 112166.67}$$

Step 5: Predict on Test Data

Area (x)	Actual Price (y)
1200	220000
1800	275000
2500	345000

- **For 1200 sqft:**

$$\hat{y} = 93 \cdot 1200 + 112166.67 = 111600 + 112166.67 = \boxed{223766.67}$$

- **For 1800 sqft:**

$$\hat{y} = 93 \cdot 1800 + 112166.67 = 167400 + 112166.67 = \boxed{279566.67}$$

- **For 2500 sqft:**

$$\hat{y} = 93 \cdot 2500 + 112166.67 = 232500 + 112166.67 = \boxed{344666.67}$$

Step 6: Calculate Mean Squared Error (MSE)

- **Training Data:**

x	Actual y	Predicted $\hat{y}$	Error	Squared Error
1000	205000	93×1000 + 112167 = 205167	-167	27,889
1500	252000	93×1500 + 112167 = 251667	333	110,889
2000	298000	93×2000 + 112167 = 298167	-167	27,889

$$\text{MSE}_{\text{train}} = \frac{27889 + 110889 + 27889}{3} = \frac{166667}{3} = \boxed{55555.67}$$

- Test Data:

x	Actual y	Predicted $\hat{y}$	Error	Squared Error
1200	220000	223767	-3767	14,189,689
1800	275000	279567	-4567	20,873,689
2500	345000	344667	333	110,889

$$\text{MSE}_{\text{test}} = \frac{14189689 + 20873689 + 110889}{3} = \frac{35,173,267}{3} = \boxed{11,724,422.33}$$

Answer

Slope (m)	93
Intercept (b)	112166.67
Model	y=93x+112166.67y = 93x + 112166.67
MSE (Train)	55,555.67
MSE (Test)	11,724,422.33

Problem3: A set of 3 points: $(-2,-1),(1,1),(3,2)$

(a) Find the **least squares regression line** for the given data points.

(b) Plot the given points and the regression line in the same rectangular system of axes.

Step 1: Calculate the values

Linear regression line:

y=a+bx

Where:

- $b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$
- $a = \frac{1}{n}\left(\sum y - b\sum x\right)$

x	y	xy	x^2
-2	-1	2	4
1	1	1	1
3	2	6	9
$\Sigma x=2$	$\Sigma y=2$	$\Sigma xy=9$	$\Sigma x^2=14$

n=3

Step 2: Calculate slope (b)

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$b = \frac{3 \cdot 9 - 2 \cdot 2}{3 \cdot 14 - 2^2} = \frac{27 - 4}{42 - 4} = \frac{23}{38}$$

Step 3: Calculate intercept (a)

$$a = \frac{1}{n}\left(\sum y - b\sum x\right) = \frac{1}{3}\left(2 - \frac{23}{38}\cdot 2\right)$$

$$= \frac{1}{3}\left(2 - \frac{46}{38}\right) = \frac{1}{3}\cdot\frac{76-46}{38} = \frac{1}{3}\cdot\frac{30}{38} = \frac{30}{114} = \frac{5}{19}$$

Regression Line:

$$y = a + bx$$

$$\boxed{y = \frac{5}{19} + \frac{23}{38}x}$$

Probably Approximately Correct (PAC) Learning

- **Probably Approximately Correct (PAC) Learning** is a framework in **machine learning theory** that describes how a learner can **efficiently learn a function** from a given class of functions, based on a limited number of training examples.

1. **Probably**: With high probability ($1 - \delta$), the learned hypothesis is good.
2. **Approximately Correct**: The hypothesis makes at most ε error on unseen examples.

A concept class C is **PAC-learnable** by a hypothesis class H if there exists a learning algorithm such that, **for any target concept** $c \in C$, and for any distribution D over the instance space, and for any $0 < \epsilon, \delta < 1$, the algorithm can find a hypothesis $h \in H$ such that:

$$P(\text{error}(h) \leq \epsilon) \geq 1 - \delta$$

Using a number of training examples **polynomial in** $\frac{1}{\epsilon}$, $\frac{1}{\delta}$, and input size.

Example

Suppose you're trying to learn whether an email is **spam** or **not spam** based on features like:

•Contains “free”

•Contains “money”

•Contains attachment

Let the true rule (target concept) be:

If the email contains both “free” and “money”, it’s spam.

You do not know the rule, but you have access to labeled data (examples).

Your hypothesis space might include simple Boolean combinations of features.

PAC learning tells us that, with **enough samples**, your learned rule will:

•Have low error (approx. correct, within ε),

•With high confidence (probably, at least $1-\delta$).

PAC Learning Formula (Sample Complexity)

To achieve **error** $\leq \epsilon$ with **probability** $\geq 1 - \delta$, you need at least:

$$m \geq \frac{1}{\epsilon}\left(\ln|H| + \ln\frac{1}{\delta}\right)$$

Where:

- m: Number of training samples
- ϵ: Acceptable error (e.g., 0.05)
- δ: Confidence gap (e.g., 0.05 means 95% confidence)
- $|H|$: Size of the hypothesis space (number of hypotheses)

- The **hypothesis space** is the set of **all possible functions** (or models) that a learning algorithm can choose from, **to approximate the true function** behind the data.

Example: Suppose a hypothesis class H has 1,000 hypotheses. How many training examples are needed to ensure that with probability at least 95%, the hypothesis has error at most 0.1?

1. $|H|=1000$
2. $\delta=0.5$
3. $\epsilon=0.1$

Solution:

Use the PAC formula:

$$m \geq \frac{1}{\epsilon}\left(\ln|H| + \ln\frac{1}{\delta}\right)$$

Plug in values:

$$m \geq \frac{1}{0.1}(\ln(1000) + \ln(20))$$

- $\ln(1000) \approx 6.908$
- $\ln(20) \approx 2.996$

$$m \geq 10 \times (6.908 + 2.996) = 10 \times 9.904 = 99.04$$

You need at least **100 training examples** to guarantee (with 95% confidence) that your hypothesis has error at most 0.1.

ε (epsilon)	Tolerable error rate
δ (delta)	Confidence shortfall (1 − δ = desired confidence)
H	Hypothesis space
m	Minimum sample size needed

PAC learning provides a **theoretical guarantee** for learning under uncertainty, ensuring models generalize well if trained on enough data.

Bayes theorem in ML

- **Bayes' Theorem** is a fundamental concept in probability theory and machine learning, especially in probabilistic models like **Naive Bayes classifiers**. It provides a way to update the probability estimate for a hypothesis as more evidence or information becomes available.

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Where:

- $P(H|E)$: **Posterior probability** – probability of hypothesis **H** given evidence **E**
- $P(E|H)$: **Likelihood** – probability of evidence **E** given hypothesis **H**
- $P(H)$: **Prior probability** – probability of hypothesis **H** before observing the evidence
- $P(E)$: **Marginal likelihood** – probability of the evidence

Example: Email Spam Classification

- Suppose you want to determine whether an email is **spam** or **not spam** based on the appearance of certain words like "offer".
- Let’s define:
- H: Email is spam
- E: The word "offer" appears in the email

Given:

- $P(H) = 0.3$ (30% of emails are spam)
- $P(\text{"offer"} | H) = 0.8$ (80% of spam emails contain the word "offer")
- $P(\text{"offer"}) = 0.4$ (40% of all emails contain "offer")

Using Bayes' Theorem:

$$P(H|\text{"offer"}) = \frac{P(\text{"offer"}|H) \cdot P(H)}{P(\text{"offer"})} = \frac{0.8 \cdot 0.3}{0.4} = \frac{0.24}{0.4} = 0.6$$

given that the word "offer" appears, there's a **60% chance** the email is spam.

Bayes' Theorem is the basis of **Naive Bayes classifiers**, which are:
•Simple yet effective
•Used for **text classification**, **spam detection**, **sentiment analysis**, etc.

A typical learning curve might look like this:

- **On the x-axis,** you have the number of training examples or data points used to train the model.
- **On the y-axis,** you typically show a performance metric, such as accuracy, error, or loss, which measures how well the model is performing on a specific task.
- The learning curve can exhibit different patterns, which can help you understand the behavior of your model:
- **Underfitting:** If the learning curve shows that the training and validation errors are both high and converge to a similar value, it suggests that the model is underfitting. This means the model is too simple to capture the underlying patterns in the data.

- **Overfitting:** When the training error is significantly lower than the validation error, it indicates overfitting. The model has learned to fit the training data very well but doesn't generalize to new, unseen data.
- **Ideal Fit:** In an ideal scenario, the training and validation errors both decrease and plateau at a relatively low level. This indicates that the model is learning the data's underlying patterns without overfitting or underfitting.
- Learning curves are crucial for determining the model's training requirements, the sufficiency of the dataset, and for making decisions on model complexity and regularization. They help machine learning practitioners optimize their models and achieve better results.

Finite and Infinite Hypothesis Spaces

- In machine learning, hypothesis spaces play a critical role in the modeling process, and they can be classified into two broad categories: finite hypothesis spaces and infinite hypothesis spaces.

- Finite Hypothesis Space:

- A finite hypothesis space is one in which the set of possible hypotheses or models is limited and countable.
- Each hypothesis in a finite hypothesis space corresponds to a specific, predefined model with a finite number of parameters.
- Training a model in a finite hypothesis space involves choosing the best hypothesis from the finite set of possibilities.
- Examples of finite hypothesis spaces include decision trees with a limited depth, k-nearest neighbors with a fixed k value, and linear regression with a fixed set of features.

- Infinite Hypothesis Space:

- An infinite hypothesis space, as the name suggests, consists of an uncountable number of possible hypotheses or models.
- These hypothesis spaces are characterized by their flexibility and ability to represent a wide range of functions.
- Learning from an infinite hypothesis space typically involves adjusting model parameters to minimize a loss function, allowing for an extensive range of possible models.
- Common examples of infinite hypothesis spaces include polynomial regression (which can represent an infinite range of polynomial functions), neural networks with varying numbers of layers and units, and support vector machines with infinite-dimensional kernel functions.

- **Key Differences:**
- Finite hypothesis spaces are more constrained, and the model complexity is limited by the predefined set of models, making them less expressive but computationally efficient.
- Infinite hypothesis spaces are more flexible and expressive, allowing models to capture complex relationships in the data but often requiring more data and computational resources.
- Finite hypothesis spaces are typically used in situations where the underlying relationships are believed to be relatively simple, and model complexity needs to be controlled.
- Infinite hypothesis spaces are employed when dealing with complex, non-linear, or high-dimensional data where a more flexible model is necessary to fit the data accurately.

- The choice between finite and infinite hypothesis spaces depends on the problem at hand, the available data, and the trade-off between model complexity and generalization performance.
- In practice, machine learning algorithms often employ regularization techniques to strike a balance between the two, allowing for complex models within certain bounds to prevent overfitting.

Probably Approximately Correct (PAC) Learning

- The Probably Approximately Correct (PAC) learning framework is a theoretical framework used in machine learning and computational learning theory to formalize the process of learning from data and make mathematical guarantees about the learning process.
- The PAC learning framework was introduced by Leslie Valiant in the late 1980s and has been instrumental in understanding the theoretical foundations of machine learning.

Here are the key concepts in PAC learning:

- **Hypothesis Class (H):** In the PAC model, the learner chooses a hypothesis from a hypothesis class (H) to describe the data. H represents the set of possible models or hypotheses that the learner considers. The learner's goal is to find a hypothesis in H that accurately describes the underlying, but unknown, data distribution.
- **Sample Complexity (m):** The sample complexity (m) is the minimum number of training examples needed for the learner to achieve a certain level of accuracy and confidence in its hypothesis. The sample complexity is typically a function of the desired error rate (ε) and confidence level (δ).

- **Approximation Error (ε):** The approximation error represents the difference between the error of the hypothesis chosen by the learner and the error of the best possible hypothesis in the hypothesis class H. The learner's goal is to keep the approximation error within a certain acceptable range.

- **Confidence Level (δ):** The confidence level (δ) determines the probability with which the learner's hypothesis should be accurate. In other words, it represents the probability that the learner's hypothesis will not be far from the true hypothesis.

- The main concept in PAC learning is that a learning algorithm is considered successful if, with high probability ($1 - \delta$), it can find a hypothesis with an error rate within ε of the best hypothesis in the chosen hypothesis class H, using a sample size of at most m training examples.

Bayes theorem in ML

- Bayes' Theorem, named after the Reverend Thomas Bayes, is a fundamental concept in probability theory and statistics that plays a crucial role in various aspects of machine learning, especially in Bayesian statistics and probabilistic modeling.
- It allows us to update our beliefs or probabilities about an event or hypothesis based on new evidence or data.
- In the context of machine learning, Bayes' Theorem is particularly useful in Bayesian inference and probabilistic modeling.

- The theorem is expressed as follows:
- Bayes' Theorem:
- $P(A|B) = (P(B|A) * P(A)) / P(B)$

- Where:
- $P(A|B)$ is the conditional probability of event A given event B.
- $P(B|A)$ is the conditional probability of event B given event A.
- $P(A)$ is the prior probability of event A.
- $P(B)$ is the marginal probability of event B.

In machine learning, Bayes' Theorem is often used in the following contexts:

- **Naive Bayes Classifiers:** In Naive Bayes classification, this theorem is used to calculate the posterior probability of a class label given a set of features. It's particularly useful for text classification, spam detection, and sentiment analysis.
- **Bayesian Inference:** In Bayesian statistics, Bayes' Theorem is used to update the prior beliefs (prior distribution) about model parameters with observed data to obtain the posterior distribution. This is especially important in probabilistic modeling and Bayesian machine learning, where you have a probabilistic description of model uncertainty.

- **Bayesian Networks:** Bayes' Theorem is central to Bayesian networks, which are graphical models that represent probabilistic relationships among variables. Inference in Bayesian networks involves using the theorem to update beliefs about variables given observed evidence.
- **Maximum A Posteriori (MAP) Estimation:** In Bayesian estimation, you use Bayes' Theorem to find the most likely parameters of a model given data. This is known as MAP estimation.
- **Bayesian Regularization:** In machine learning, Bayesian regularization (e.g., Bayesian linear regression) incorporates prior beliefs about model parameters into the learning process to improve model generalization and handle overfitting.

- Bayes' Theorem provides a principled way to incorporate prior knowledge or beliefs into the modeling process and to update those beliefs with data. It is a cornerstone of Bayesian statistics and machine learning, helping to make models more robust and adaptive to new information.

MDL principle

- The Minimum Description Length (MDL) principle is a fundamental concept in machine learning, information theory, and statistics. It is used to guide model selection, model complexity, and model evaluation by finding the simplest and most informative representation of data. The MDL principle is based on the idea that the best model for a given dataset is the one that can compress or describe the data most efficiently. In other words, it aims to strike a balance between model complexity and the ability to explain the data.

An overview of how the MDL principle is applied in machine learning:

- **Model Selection:** When faced with multiple candidate models (hypotheses), the MDL principle suggests choosing the model that minimizes the total length of the description of the data and the model. The model description should include the model parameters and structure, as well as any additional information needed to reproduce the data.

- **Model Complexity:** The MDL principle encourages choosing models with a simpler structure and fewer parameters over more complex models if both can explain the data similarly well. Simplicity is favored because simpler models have shorter descriptions, which means they are more efficient in terms of coding length.

- **Model Evaluation:** In the context of model evaluation, the MDL principle provides a basis for comparing different models. The model with the shortest description length (i.e., that can best compress the data) is often considered the best model for the given data.
- **Bayesian Interpretation:** In a Bayesian framework, the MDL principle can be seen as equivalent to selecting models that maximize the posterior probability given the data. In other words, it suggests selecting the most probable model given the data and prior beliefs.
- **Occam's Razor:** The MDL principle is closely related to Occam's razor, which is a philosophical principle stating that among competing hypotheses, the simplest one should be preferred. The MDL principle operationalizes this by quantifying the simplicity of a model through the length of its description.

- The MDL principle is a powerful tool for addressing the bias-variance trade-off in machine learning.
- It discourages overly complex models that may overfit the data and favors simpler models that are more likely to generalize well to unseen data.
- However, it's important to note that implementing the MDL principle in practice can be challenging, as it often involves complex model selection and encoding schemes.
- Nevertheless, it remains a valuable theoretical framework for understanding and guiding model selection and evaluation in machine learning.

THANK YOU!!!